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Module 3 Basics of Deep Learning and Foundation Models

Module 3.1: Perceptron — 20 Questions

Single Choice Questions (1-7)

Q1 The material states that in 1957, Frank Rosenblatt at Cornell University invented the perceptron. Which of the following statements about the original perceptron is CORRECT?

- A. It was a multi-layer neural network with backpropagation
- B. It was a binary linear classifier using the sign function as activation
- C. It could solve non-linear problems like XOR from the beginning
- D. It was implemented only in software and never built as hardware

Q2 The material describes the perceptron mathematical representation:

$$\begin{cases} -1 & \text{if } w_1x_1 + w_2x_2 \leq \theta \\ 1 & \text{if } w_1x_1 + w_2x_2 > \theta \end{cases}$$

$$y = f(x) = \begin{cases} -1 & \text{if } w_1x_1 + w_2x_2 \leq \theta \\ 1 & \text{if } w_1x_1 + w_2x_2 > \theta \end{cases}$$

$$\begin{cases} -1 & \text{if } w_1x_1 + w_2x_2 \leq \theta \\ 1 & \text{if } w_1x_1 + w_2x_2 > \theta \end{cases}$$

A perceptron receives inputs $x_1 = 0.8$, $x_2 = 0.3$ with weights $w_1 = 0.5$, $w_2 = 0.5$, and threshold $\theta = 0.7$.

What is the output y ?

- A. -1
- B. 0
- C. 1
- D. 0.55

Q3 The material shows that an AND gate can be implemented with $(w_1, w_2, \theta) = (0.5, 0.5, 0.8)$.

Which weight-threshold combination implements a NAND gate?

- A. (0.5, 0.5, 0.8)
- B. (-0.5, -0.5, -0.7)
- C. (0.5, 0.5, 0.3)
- D. (1.0, 1.0, 1.5)

Q4 The material asks: "Can we use a single-layer perceptron to distinguish between an XOR gate?"

What is the correct answer and the fundamental reason?

- A. Yes, because XOR is a linearly separable problem
- B. No, because XOR is not linearly separable - a single straight line cannot separate the outputs
- C. Yes, but only with a very large threshold value
- D. No, because XOR requires more than two inputs

Q5 The material shows that XOR can be implemented by combining AND, NAND, and OR gates in a two-layer perceptron.

In the XOR implementation circuit shown, what should replace the three question marks (?) from left to right?

- A. NAND, OR, AND
- B. AND, OR, NAND
- C. OR, NAND, AND
- D. NAND, AND, OR

Q6 The material explains two sources of nonlinearity in multi-layer perceptrons: "More layers" and "Nonlinear activation function."

If we remove the sign function from the middle layer of a 3-layer perceptron and use only linear operations, what happens?

- A. The network becomes more powerful because it has fewer constraints
- B. The network remains equivalent to a single linear classifier, losing its nonlinearity
- C. The network can still solve XOR because of the multiple layers
- D. The network becomes a deep neural network automatically

Q7 The material states: "perceptron signals form a flow and transmit information forward. The difference from an electric current is that a perceptron signal has only two values, current (1) and no current (0)."

In modern neural networks, how has this signal representation evolved?

- A. Signals are still strictly binary (0 or 1)
- B. Signals can be continuous real values after activation functions
- C. Signals are always negative
- D. Signals are eliminated entirely in deep networks

Multiple Choice Questions (8-11) [⚠](#) (Select ALL that apply)

Q8 Which statements about the perceptron are CORRECT according to the material? (Select all that apply)

- A. The perceptron is considered the simplest form of a feedforward neural network
- B. Weights determine the importance of each input signal
- C. A larger weight indicates that the signal is more important
- D. The perceptron can solve any classification problem regardless of complexity
- E. The threshold θ indicates the difficulty level to activate the neuron

Q9 The material discusses the XOR problem extensively. Which statements are CORRECT? (Select all that apply)

- A. OR gate is linearly separable and can be solved by a single-layer perceptron
- B. XOR gate is NOT linearly separable in 2D space
- C. A two-layer perceptron can solve XOR by combining AND, NAND, and OR gates
- D. The XOR truth table outputs 1 only when both inputs are 1
- E. A non-linear curve is required to separate XOR outputs

Q10 The material shows a 3-layer perceptron with input layer, middle layer, and output layer. Which statements are CORRECT? (Select all that apply)

- A. The input layer receives signals and sends them to the middle layer
- B. The middle layer processes signals using the sign function and sends to the output layer
- C. The output layer produces the final result after processing
- D. Without nonlinear activation functions, multiple layers are mathematically equivalent to a single layer
- E. Adding more layers always improves performance regardless of activation functions

Q11 The material contrasts traditional machine learning with the question: "Can you use an algorithm to automatically execute the procedure?"

Which statements describe the key differences? (Select all that apply)

- A. Traditional ML requires manual feature extraction and selection
- B. Deep learning (neural networks) can automatically learn feature extraction
- C. Traditional ML uses data cleansing, feature extraction, and feature selection as separate manual steps
- D. Neural networks eliminate the need for data entirely
- E. More features in traditional ML always improve performance without any drawbacks

True or False Questions (12-15)

Q12 The perceptron was invented by Frank Rosenblatt in 1957 and is considered the first practical application of artificial neural networks.

- A. True
- B. False

Q13 In the perceptron, weights are similar to resistance in an electric circuit - a larger weight indicates a smaller signal that passes through.

- A. True
- B. False

Q14 A single-layer perceptron with sign function can solve the XOR problem if we adjust the weights and threshold appropriately.

- A. True
- B. False

Q15 The material states that in a multi-layer perceptron, if the sign function is not used at the middle layer, the outputs a_1 and a_2 remain linear combinations of inputs, and the final output y is still a linear function of the original inputs.

- A. True
- B. False

Fill in the Blank Questions (16-18)

Q16 A perceptron receives multiple _____ signals and outputs one signal. The input signals are multiplied by fixed _____ (w_1x_1 and w_2x_2). The neuron calculates the _____ of the input signals and outputs 1 only when the sum exceeds a certain _____.

Q17 The XOR gate outputs 1 when _____ x_1 or x_2 is 1, but NOT when _____. In the 2D coordinate system, the XOR outputs cannot be separated by a _____, which is why a _____-layer perceptron is needed.

Q18 The two sources of nonlinearity in multi-layer perceptrons are: (1) _____, and (2) _____. The sign function determines whether the neuron is _____ based on the output in the previous phase, and is therefore called the _____.

Drag and Drop / Matching Questions (19-20)

Q19 Match each logic gate to its property:

Logic Gate	Property
A. AND gate	1. Outputs 1 when either input is 1 (but not both)
B. OR gate	2. Outputs 1 only when both inputs are 1
C. NAND gate	3. Outputs 0 only when both inputs are 1
D. XOR gate	4. Outputs 1 when at least one input is 1

Write your answer as: A→?, B→?, C→?, D→?

Q20 Match each perceptron component to its function:

Component	Function
A. Weight (w)	1. Determines if neuron activates based on weighted sum
B. Threshold (θ)	2. Controls the importance of each input signal
C. Sign function	3. Introduces nonlinearity to enable complex pattern recognition
D. Multiple layers	4. Sets the minimum weighted sum required for activation

Write your answer as: A→?, B→?, C→?, D→?

Q	Answer	Explanation
1	B	Material explicitly states: "binary linear classifier...activation function is a sign function"
2	C	$w_1x_1 + w_2x_2 = 0.5 \times 0.8 + 0.5 \times 0.3 = 0.4 + 0.15 = 0.55$. $0.55 \leq 0.7$? No, $0.55 < 0.7$, so output = -1... Wait, let me recalculate: $0.55 \leq 0.7$ is TRUE, so $y = -1$. Actually: $0.55 \leq 0.7 \rightarrow -1$. But wait, 0.55 is NOT > 0.7 , so it's -1. Hmm, but let me check again: $w_1x_1 + w_2x_2 = 0.5 \times 0.8 + 0.5 \times 0.3 = 0.4 + 0.15 = 0.55$. $0.55 \leq 0.7 \rightarrow y = -1$. So answer is A. But wait, the formula says -1 when $\leq \theta$, and 1 when $> \theta$. $0.55 \leq 0.7$, so $y = -1$. Answer: A
3	B	NAND is the inverse of AND. With negative weights and negative threshold, it inverts the AND behavior
4	B	Material explicitly shows XOR is not linearly separable - cannot divide with a straight line
5	D	$XOR = (x_1 \text{ NAND } x_2) \text{ AND } (x_1 \text{ OR } x_2)$. The circuit shows: top gate = NAND, bottom = AND, final = OR... Actually from material: $XOR = NAND \rightarrow AND \rightarrow OR$? Let me check the truth table implementation. The material shows XOR gate implementation using AND, NAND, OR. The standard implementation is: $s_1 = x_1 \text{ NAND } x_2$, $s_2 = x_1 \text{ OR } x_2$, output = $s_1 \text{ AND } s_2$. So left = NAND, middle = OR, right = AND. Wait, looking at the diagram again: two inputs go to two gates in parallel, then to one final gate. The standard is: NAND (top), OR (bottom), AND (final). So answer should be D (NAND, AND, OR) if reading left-to-right as NAND, AND, OR... Actually let me reconsider. The diagram shows: inputs split to two parallel gates (marked ?), then their outputs go to a final gate (marked ?). Standard XOR: $s_1 = \text{NAND}(x_1, x_2)$, $s_2 = \text{OR}(x_1, x_2)$, $y = \text{AND}(s_1, s_2)$. So left-to-right: NAND, OR, AND. But that's not in the options... Wait, looking at options: A. NAND, OR, AND - that's actually correct! Let me re-read the options. A is NAND, OR, AND. Yes! Answer: A
6	B	Material explicitly shows: without sign function, a_1 and a_2 are linear, and final output is linear combination of linear = still linear
7	B	Modern neural networks use continuous activation functions (sigmoid, ReLU, etc.) producing real-valued outputs
8	A, B, C, E	D is false - perceptron cannot solve all problems (e.g., XOR)
9	A, B, C, E	D is false - XOR outputs 1 when EITHER (not both) input is 1
10	A, B, C, D	E is false - without nonlinear activation, more layers don't help
11	A, B, C	D is false - neural networks still need data; E is false - more features increase computation and can cause issues
12	A - True	Direct quote from material
13	B - False	Material states: "a larger weight indicates a larger signal that passes through" - opposite of resistance
14	B - False	Material explicitly proves XOR cannot be solved by single-layer perceptron
15	A - True	Material shows the mathematical proof that without sign function, output remains linear
16	input, weights, sum, threshold	From perceptron description
17	either, both are 1, straight line, two	From XOR problem description
18	More layers, Nonlinear activation function, activated, activation function	From nonlinearity sources section
19	A→2, B→4, C→3, D→1	AND=both 1, OR=at least one 1, NAND=NOT AND, XOR=either but not both
20	A→2, B→4, C→1, D→3	Weight=importance, Threshold=minimum sum, Sign=activation decision, Layers=nonlinearity/complexity

Module 3.2: Fully Connected Neural Network & Activation Functions — 30 Questions

Single Choice Questions (1-10)

Q1 A Huawei AI team is building a deep neural network for image classification. They start with a simple perceptron (1 middle layer with 2 units) that can only solve binary classification. To handle 10 different object categories with higher accuracy, they increase both the number of hidden layers and nodes per layer.

According to the deep neural network architecture, what is the PRIMARY benefit of adding more hidden layers?

- A. It reduces the total number of parameters in the model
- B. It increases the network's identification capability and ability to fit complex functions
- C. It eliminates the need for activation functions
- D. It makes the network require less training data

Q2 A feedforward neural network (FNN) has the following structure: Input layer → Hidden Layer 1 → Hidden Layer 2 → Output layer. Each neuron in Hidden Layer 1 receives weighted sums from ALL input neurons, applies an activation function, and passes output to Hidden Layer 2.

What does this architecture demonstrate?

- A. Recurrent connections with feedback loops
- B. Fully-connected layers with unidirectional signal transmission
- C. Convolutional operations with local receptive fields
- D. Sparse connections to reduce computation

Q3 A neural network for binary classification outputs values that need to be interpreted as probabilities (between 0 and 1). The output layer should use:

- A. ReLU function
- B. Sigmoid function
- C. Step function
- D. Linear function

Q4 A deep neural network with 20 hidden layers uses sigmoid activation functions throughout. During training, the gradients in the early layers become extremely small, causing those layers to barely update.

What is this phenomenon called, and what is the root cause?

- A. Exploding gradient — caused by weights greater than 1
- B. Vanishing gradient — caused by sigmoid derivative maximum being 0.25 and chain rule multiplication
- C. Overfitting — caused by too much training data
- D. Underfitting — caused by insufficient model capacity

Q5 A convolutional neural network for image recognition uses ReLU as the activation function. Compared to sigmoid, what is the PRIMARY advantage of ReLU in deep networks?

- A. ReLU outputs are bounded between 0 and 1
- B. ReLU derivative is 1 for positive inputs, preventing vanishing gradients
- C. ReLU is centrosymmetric about the origin
- D. ReLU requires more computation than sigmoid

Q6 The Swish function is defined as: $\text{swish}(x) = x \cdot \sigma(\beta x)$, where σ is the sigmoid function.

When $\beta \rightarrow +\infty$, what does the Swish function approximate?

- A. A linear function $f(x) = x/2$
- B. The ReLU function
- C. The tanh function
- D. A constant function

Q7 A neural network for sentiment analysis needs an activation function that is centrosymmetric about the origin (output range centered around 0). Which function should be chosen?

- A. Sigmoid — outputs (0, 1)
- B. Tanh — outputs (-1, 1) and is centrosymmetric about origin
- C. ReLU — outputs $[0, +\infty)$
- D. Step function — outputs $\{-1, 1\}$ only

Q8 Standard ReLU sets all negative values to exactly 0. In a deep network, some neurons might always output negative values and never activate, effectively "dying."

What problem does Leaky ReLU solve?

- A. It prevents the dying ReLU problem by allowing small negative slopes
- B. It makes the function linear for all inputs
- C. It bounds the output between -1 and 1
- D. It eliminates the need for bias terms

Q9 A neural network classifies Huawei cloud service tickets into 5 categories: Billing, Technical, Account, Security, and General. The output layer produces a 5-dimensional vector.

What properties does the Softmax output have?

- A. Each element is in (-1, 1) and sum equals 0
- B. Each element is in (0, 1) and sum equals 1, interpretable as probabilities
- C. Each element is exactly 0 or 1
- D. The maximum element is always 1 and others are 0

Q10 A data scientist proposes using $f(x) = wx + b$ (linear activation) for ALL layers in a 10-layer deep neural network.

What is the fundamental problem with this approach?

- A. The network will have too many parameters
- B. The entire network is mathematically equivalent to a single linear layer, defeating the purpose of depth
- C. Linear activations cause vanishing gradients
- D. The network cannot use backpropagation

Multiple Choice Questions (11-16)  (Select ALL that apply)

Q11 Which statements about Feedforward Neural Networks (FNN) are CORRECT? (Select all that apply)

- A. The first layer is the input layer providing initial data
- B. Each neuron at hidden and output layers receives weighted sums from all neurons in the previous layer
- C. Signals are transmitted unidirectionally from input to output
- D. The network can be represented as a directed acyclic graph
- E. Feedback connections from output to input are essential for learning

Q12 Which statements about common activation functions are CORRECT? (Select all that apply)

- A. Sigmoid maps any real input to (0, 1) and is called a squeezing function
- B. Tanh maps inputs to (-1, 1) and is centrosymmetric about the origin
- C. ReLU outputs $\max(0, x)$ and is one of the most popular activation functions
- D. Step function has a derivative of 0 almost everywhere
- E. Softmax is used for multi-class classification and outputs a probability distribution

Q13 A deep neural network with many hidden layers faces several challenges. Which statements are CORRECT? (Select all that apply)

- A. More hidden layers generally increase identification capability
- B. Simply adding nodes to a single layer cannot match the benefits of adding multiple layers
- C. Deep networks still require nonlinear activation functions to be effective
- D. Without nonlinear activations, multiple layers collapse to a single linear transformation
- E. Adding 20 hidden layers always guarantees better performance than 3 hidden layers

Q14 The Swish function $\text{swish}(x) = x \cdot \sigma(\beta x)$ has interesting properties at different β values. Which statements are CORRECT? (Select all that apply)

- A. When $\beta = 0$, Swish becomes a linear function $x/2$
- B. When $\beta = 1$, Swish is approximately linear for $x > 0$ and non-monotonic for $x < 0$
- C. When $\beta \rightarrow +\infty$, Swish approximates the ReLU function
- D. Swish can be considered a nonlinear interpolation between linear and ReLU
- E. Swish is always monotonically increasing regardless of β value

Q15

In a fully-connected layer, a neuron computes: $a = \sum(x_i \cdot w_i) + b$, then $y = f(a)$, where f is the activation function. Which statements about this computation are CORRECT? (Select all that apply)

- A. x_i represents input values from the previous layer
- B. w_i represents weights connecting inputs to the current neuron
- C. b is the bias term (threshold) for activating the neuron
- D. $f(a)$ is the nonlinear activation function
- E. The output y is always a binary value (0 or 1)

Q16 Match the following scenarios to appropriate activation functions. Which combinations are CORRECT? (Select all that apply)

- A. Binary classification output layer $\rightarrow$ Sigmoid
- B. Multi-class classification output layer $\rightarrow$ Softmax
- C. Hidden layers in deep CNN $\rightarrow$ ReLU
- D. When outputs need to be centered around 0 $\rightarrow$ Tanh
- E. When a simple threshold is sufficient $\rightarrow$ Step function

True or False Questions (17-20)

Q17 Deep learning is a subset of machine learning, and the deep learning architecture is generally a deep neural network where "depth" refers to the number of layers.

- A. True
- B. False

Q18 The step function (sign function) used in perceptrons is ideal for modern deep neural networks because its derivative is well-defined and nonzero everywhere, enabling smooth gradient-based optimization.

- A. True
- B. False

Q19 In a feedforward neural network, if linear activation $f(x) = wx + b$ is used in all layers, a 10-layer network with weights $W_1, W_2, \dots, W_{10}$ is mathematically equivalent to a single layer with weight matrix $W = W_{10} \cdot W_9 \cdot \dots \cdot W_2 \cdot W_1$.

- A. True
- B. False

Q20 The sigmoid function compresses any input in the range $(-\infty, +\infty)$ to a value within $(0, 1)$, which is why it is often referred to as a squeezing function. However, when inputs are very large positive or negative, the sigmoid gradient approaches zero, contributing to the vanishing gradient problem.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q 21 A Feedforward Neural Network (FNN) has three main components: the _____ layer provides initial data, the _____ layer(s) process features, and the _____ layer produces final results. The network uses a _____ multi-layer structure with no feedback.

Q22 The sigmoid function is defined as $\sigma(x) = \frac{1}{1 + e^{-x}}$. It maps inputs to the interval _____ and is called a _____ function. Its derivative maximum value is _____, which contributes to the _____ problem in deep networks.

Q23 ReLU is defined as $\max(0, x)$. Its derivative for positive inputs is _____, which helps prevent _____ . Leaky ReLU improves upon ReLU by assigning a _____ slope to negative values, solving the _____ problem.

Q24 The Swish function $\text{swish}(x) = x \cdot \sigma(x)$ is a _____-gated activation function. When $\beta = 0$, it becomes _____. When $\beta \rightarrow +\infty$, it approximates _____. It can be viewed as a nonlinear _____ between a linear function and ReLU.

Drag and Drop / Matching Questions (25-30)

Q25 Match each activation function to its output range:

Activation Function	Output Range
---------------------	--------------

- | | |
|------------|--------------------------|
| A. Sigmoid | 1. $(-1, 1)$ |
| B. Tanh | 2. $(0, 1)$ |
| C. ReLU | 3. $[0, +\infty)$ |
| D. Softmax | 4. $(0, 1)$ with sum = 1 |

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each deep learning problem to its solution:

Problem	Solution
---------	----------

- | | |
|--|--------------------------|
| A. Vanishing gradient with sigmoid | 1. Use gradient clipping |
| B. Exploding gradient | 2. Use ReLU or LSTM |
| C. Dying ReLU | 3. Use Leaky ReLU |
| D. Need probability distribution for multi-class | 4. Use Softmax |

Write your answer as: A→?, B→?, C→?, D→?

Q27 Match each network configuration to its classification capability:

Configuration	Capability
---------------	------------

- | | |
|--|--|
| A. 0 hidden layers | 1. Can fit highly complex non-linear functions |
| B. 3 hidden layers | 2. Linear classification only |
| C. 20 hidden layers with nonlinear activations | 3. Moderate non-linear function fitting |

Write your answer as: A→?, B→?, C→?

Q28 Match each term to its definition in FNN:

Term Definition

- | | |
|------------------------|---|
| A. Weight (W) | 1. Threshold for neuron activation |
| B. Bias (B) | 2. Strength of connection between neurons |
| C. Activation function | 3. Input data source |
| D. Input layer | 4. Nonlinear transformation applied to weighted sum |

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each scenario to the most appropriate activation function:

Scenario Activation

- | | |
|--|------------|
| A. Hidden layers in deep CNN for image recognition | 1. Sigmoid |
| B. Output layer for binary spam detection | 2. ReLU |
| C. Output layer for 10-digit handwriting recognition | 3. Softmax |
| D. When outputs should be centered around zero | 4. Tanh |

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each β value to Swish function behavior:

β Value Behavior

- | | |
|--------------------------------|---|
| A. $\beta = 0$ | 1. Approximates ReLU |
| B. $\beta = 1$ | 2. Linear function $x/2$ |
| C. $\beta \rightarrow +\infty$ | 3. Non-monotonic for $x < 0$, linear for $x > 0$ |

Write your answer as: A→?, B→?, C→?

Q	Answer	Explanation
1	B	More hidden layers increase identification capability and ability to fit complex functions
2	B	FNN has fully-connected layers with unidirectional signal transmission (no feedback)
3	B	Sigmoid outputs (0, 1), interpretable as probability for binary classification
4	B	Vanishing gradient: sigmoid derivative max = 0.25, chain rule makes gradients vanish in deep networks
5	B	ReLU derivative = 1 for $x > 0$, preventing vanishing gradients
6	B	When $\beta \rightarrow +\infty$, $\sigma(\beta x)$ becomes step function, so $\text{swish}(x) \approx x \cdot \text{step}(x) = \text{ReLU}(x)$
7	B	Tanh is centrosymmetric about origin, outputs (-1, 1)
8	A	Leaky ReLU allows small negative slope (αx), preventing dying ReLU
9	B	Softmax outputs (0, 1) with sum = 1, interpretable as probabilities
10	B	Linear + Linear = Linear. Multiple linear layers collapse to single linear transformation
11	A, B, C, D	E is false - FNN has no feedback connections
12	A, B, C, D, E	All statements are correct
13	A, B, C, D	E is false - more layers don't always guarantee better performance (can overfit or have optimization issues)
14	A, B, C, D	E is false - Swish is non-monotonic for some β values ($x < 0$)
15	A, B, C, D	E is false - output y depends on activation function, not always binary
16	A, B, C, D, E	All combinations are correct
17	A - True	Direct definition from the content
18	B - False	Step function derivative is 0 almost everywhere (except at 0), making gradient-based learning impossible
19	A - True	Linear transformations are closed under composition: $W_{\text{total}} = W_{10} \cdot W_9 \cdot \dots \cdot W_1$
20	A - True	Sigmoid saturates at extremes, gradient $\rightarrow 0$, causing vanishing gradients
21	input, hidden, output, unidirectional	FNN structure description
22	$1/(1+\exp(-x))$, (0, 1), squeezing, 0.25, vanishing gradient	Sigmoid properties
23	$\max(0, x)$, 1, vanishing gradient, non-zero, dying ReLU	ReLU and Leaky ReLU properties
24	$x \cdot \sigma(\beta x)$, self, $x/2$, ReLU, interpolation	Swish function properties
25	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	Sigmoid=(0,1), Tanh=(-1,1), ReLU= $\setminus[0, \infty)$, Softmax=prob distribution
26	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	Vanishing $\rightarrow$ ReLU/LSTM, Exploding $\rightarrow$ clipping, Dying ReLU $\rightarrow$ Leaky ReLU, Multi-class $\rightarrow$ Softmax
27	A $\rightarrow$ 2, B $\rightarrow$ 3, C $\rightarrow$ 1	0 layers=linear, 3 layers=moderate, 20 layers=complex non-linear
28	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 3	Weight=connection strength, Bias=threshold, Activation=nonlinear transform, Input layer=data source
29	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	CNN hidden=ReLU, Binary output=Sigmoid, Multi-class=Softmax, Centered=Tanh
30	A $\rightarrow$ 2, B $\rightarrow$ 3, C $\rightarrow$ 1	$\beta=0 \rightarrow x/2$, $\beta=1 \rightarrow$ non-monotonic/linear, $\beta \rightarrow \infty \rightarrow \text{ReLU}$

Module 3.2.3: Neural Network Training — 30 Questions

Single Choice Questions (1-10)

Q1 A Huawei AI team is training a neural network to classify 1 million product images into 500 categories. They need to automatically learn features from raw pixel data rather than manually designing features like SIFT or HOG.

Which approach best describes their strategy?

- A. Traditional machine learning with manual feature engineering
- B. Deep learning where the neural network automatically learns feature extraction
- C. Rule-based classification with explicit programming
- D. Unsupervised clustering without any labels

Q2 A dataset of 100,000 medical images is divided into training, validation, and test sets. The team notices that after 50 epochs, the training loss continues to decrease but the validation loss starts increasing.

What does this indicate, and which set should be used to detect this problem?

- A. Underfitting detected by the training set
- B. Overfitting detected by the validation set
- C. Good generalization detected by the test set
- D. Data leakage detected by the training set

Q3 A neural network predicts house prices in Shenzhen based on features like area, location, and floor level. The output is a continuous numerical value.

Which loss function is MOST appropriate?

- A. Cross-entropy loss
- B. Mean square error (MSE)
- C. Hinge loss
- D. Binary cross-entropy

Q4 The gradient of a multivariate function $C(W)$ at point W' is:

$$\nabla f(W') = \left[\frac{\partial f}{\partial w_0}, \frac{\partial f}{\partial w_1}, \dots, \frac{\partial f}{\partial w_n} \right]^T$$

Why do we update parameters in the REVERSE gradient direction?

- A. Because the gradient points in the direction of fastest increase, and we want to minimize the loss
- B. Because the gradient points in the direction of fastest decrease
- C. Because reverse direction is computationally cheaper
- D. Because the gradient is always negative

Q5 Batch Gradient Descent (BGD) uses the entire training set for each weight update.

What is the PRIMARY advantage of BGD?

- A. Fastest convergence per iteration
- B. Each update is made towards the correct direction, ensuring convergence to extrema
- C. Lowest memory usage
- D. Best handling of noisy data

Q6 Stochastic Gradient Descent (SGD) updates weights based on a single randomly selected sample.

What is the MAIN drawback of SGD compared to BGD?

- A. It requires more memory
- B. The gradient direction fluctuates near the extremum, making convergence difficult
- C. It always converges to the global minimum
- D. It cannot handle large datasets

Q7 Mini-batch Gradient Descent (MBGD) uses a small fixed batch size (e.g., 32, 64, 128).

Why is MBGD the most commonly used gradient descent algorithm in practice?

- A. It uses the entire dataset like BGD for perfect convergence
- B. It balances the efficiency of SGD with the gradient stability of BGD
- C. It eliminates the need for a learning rate
- D. It guarantees convergence to the global minimum

Q8 In backpropagation, to compute $\frac{\partial C}{\partial w_1}$, we use: $\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_1}$

What mathematical principle enables this decomposition?

- A. Matrix multiplication associativity
- B. The chain rule for composite functions
- C. Gradient clipping
- D. The central limit theorem

Q9 In a deep network with sigmoid activations, the maximum value of $\sigma'(x)$ is 0.25. If $|w| < 1$, then $|\sigma'(z)w| \leq \frac{1}{4}$. What happens to $\frac{\partial C}{\partial w_i}$ as the number of layers increases?

- A. It increases exponentially
- B. It becomes smaller, resulting in gradient vanishing
- C. It remains constant
- D. It oscillates between positive and negative

Q10 during training (gradients become extremely large).

Which solution is MOST appropriate?

- A. Replace all activations with sigmoid
- B. Use gradient clipping to limit gradient magnitude
- C. Remove half the layers
- D. Increase the learning rate by 10x

Multiple Choice Questions (11-16)  (Select ALL that apply)

Q11A dataset is divided into training, validation, and test sets. Which statements are CORRECT? (Select all that apply)

- A. The training set is used for feature learning and model parameter optimization
- B. The validation set evaluates model training at the end of a phase for adjustments
- C. The test set reflects the model's final performance on unseen data
- D. The test set should be used frequently during training to guide adjustments
- E. The validation set is used to calculate the final model accuracy

Q12 Which statements about loss functions are CORRECT? (Select all that apply)

- A. Mean square error (MSE) is mainly used for regression tasks
- B. Cross-entropy loss is mainly used for classification tasks
- C. MSE formula : $E = \frac{1}{2} \sum_k (y_k - t_k)^2$ where y_k is network output and t_k is target
- D. Cross-entropy formula $E = - \sum_k t_k \log y_k$ where t_k is the correct label

Q13 Compare BGD, SGD, and MBGD. Which statements are CORRECT? (Select all that apply)

- A. BGD uses the entire training set for each update and converges to extrema
- B. SGD updates based on each sample, causing gradient fluctuation near extremum
- C. MBGD uses a small batch and balances efficiency and stability
- D. BGD requires the least memory resources
- E. MBGD may easily overshoot the local minimum

Q14 Which statements about the network training process are CORRECT? (Select all that apply)

- A. Forward propagation computes the output and loss from input to output
- B. Backpropagation computes gradients from output to input using the chain rule
- C. Parameters are updated as $W_{t+1} = W_t - \eta \frac{dC}{dW}$
- D. Backpropagation multiplies the loss signal by local derivatives at each node
- E. Forward and backward propagation are performed only once per training

Q15 Which statements about gradient problems in deep networks are CORRECT? (Select all that apply)

- A. Vanishing gradient: derivative value decreases during backward propagation
- B. Exploding gradient: derivative value increases during backward propagation
- C. Sigmoid's maximum derivative (0.25) contributes to vanishing gradients
- D. When $|\sigma'(z) \cdot w| > 1$, gradients may explode
- E. Gradient clipping prevents exploding by setting a threshold

Q16 Which are VALID solutions to gradient problems? (Select all that apply)

- A. Use ReLU activation function to mitigate vanishing gradients
- B. Use LSTM networks to mitigate vanishing gradients
- C. Use gradient clipping to prevent exploding gradients
- D. Use weight regularization (L1/L2) to prevent exploding gradients
- E. Use sigmoid activation in all layers to prevent both problems

True or False Questions (17-20)

Q17 The primary difference between a neural network and a multi-layer perceptron is that neural networks can automatically learn feature extraction, while perceptrons require manual feature design.

- A. True
- B. False

Q18 In gradient descent, the gradient points in the direction where the multivariate function value increases the quickest, so parameters should be optimized in the reverse gradient direction to minimize the loss.

- A. True
- B. False

Q19 BGD is not used often because convergence is very slow since all training samples need to be computed every time the weight is updated. However, each update is made towards the correct direction, ensuring convergence to extrema.

- A. True
- B. False

Q20 Gradient exploding and vanishing are caused by large network depth and unstable network weight update. In essence, they are caused by the chain rule in gradient backward propagation.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 A neural network learns by: (1) _____ to compute predictions, (2) calculating the _____ between predictions and true values using a _____, (3) _____ to compute gradients, and (4) _____ to update weights.

Q22 Batch Gradient Descent uses the _____ training set for each update. Stochastic Gradient Descent uses _____ sample for each update. Mini-batch Gradient Descent uses a _____ of samples (batch size) for each update and is the most _____ used in practice.

Q23 The chain rule states that for a composite function $z = f(g(x))$, the derivative $\frac{dz}{dx} =$ _____.

In backpropagation, this allows computing $\frac{\partial C}{\partial w_1}$ as the _____ of partial derivatives along the path from _____ to _____.

Q24 The maximum value of the sigmoid derivative $\sigma'(x)$ is _____. When $|w| < 1$, $|\sigma'(z) \cdot w| \leq$ _____. As network depth increases, the product of many terms $\leq 1/4$ causes _____. Solutions include using _____ activation or _____ networks.

Drag and Drop / Matching Questions (25-30)

Q25 Match each dataset split to its purpose:

Dataset Split Purpose

- | | |
|-------------------|---|
| A. Training set | 1. Evaluate final performance on completely unseen data |
| B. Validation set | 2. Learn features and optimize model parameters |
| C. Test set | 3. Make adjustments based on training phase results |

Write your answer as: A→?, B→?, C→?

Q26 Match each loss function to its primary task:

Loss Function Task

- | | |
|----------------------------|---------------------------------------|
| A. Mean Square Error (MSE) | 1. Multi-class classification |
| B. Cross-entropy | 2. Regression with continuous outputs |
| C. Hinge loss | 3. Binary classification with SVM |

Write your answer as: A→?, B→?, C→?

Q27 Match each gradient descent type to its key characteristic:

GD Type Characteristic

- | | |
|---------|--|
| A. BGD | 1. Fast per iteration but noisy, hard to converge |
| B. SGD | 2. Balances efficiency and stability, most commonly used |
| C. MBGD | 3. Slow but stable, guaranteed convergence direction |

Write your answer as: A→?, B→?, C→?

Q28 Match each gradient problem to its root cause:

Problem Cause

- | | |
|-------------------------|---|
| A. Vanishing gradient | 1. $ \sigma'(z) \cdot w > 1$ with many layers |
| B. Exploding gradient | 2. $\sigma'(x) \max = 0.25$, $ w < 1$, chain rule multiplication |
| C. Slow BGD convergence | 3. Entire dataset computed for each update |

Write your answer as: A→?, B→?, C→?

Q29 Match each solution to the problem it solves:

Solution Problem

- | | |
|----------------------|---|
| A. Gradient clipping | 1. Vanishing gradients in deep networks |
| B. ReLU activation | 2. Exploding gradients during training |
| C. LSTM networks | 3. Vanishing gradients in sequence models |

Write your answer as: A→?, B→?, C→?

Q30 Arrange the following neural network training steps in the correct order:

Table

Step Description

- | | |
|------------------------|---|
| A. Compute loss | 1. $W_{t+1} = W_t - \eta \frac{dC}{dW}$ |
| B. Forward propagation | 2. Calculate difference between prediction and target |
| C. Backpropagation | 3. Pass input through network to get output |
| D. Update parameters | 4. Compute gradients using chain rule |

Write your answer as: Correct sequence (e.g., B→A→C→D)

 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	B	Neural networks automatically learn feature extraction from raw data
2	B	Training loss ↓ + validation loss ↑ = overfitting, detected by validation set
3	B	MSE is for regression tasks with continuous outputs
4	A	Gradient points to fastest increase; reverse direction minimizes loss
5	B	BGD uses entire dataset → correct direction → guaranteed convergence
6	B	SGD's single-sample updates cause fluctuation near extremum
7	B	MBGD balances SGD efficiency + BGD stability
8	B	Chain rule for composite functions enables gradient decomposition
9	B	Product of terms $\leq 1/4$ becomes very small → vanishing gradient
10	B	Gradient clipping limits gradient magnitude to prevent explosion
11	A, B, C	D is false (test set used only once); E is false (validation not for final accuracy)
12	A, B, C, D	E is false (different loss functions for different tasks)
13	A, B, C, E	D is false (BGD requires MOST memory, not least)
14	A, B, C, D	E is false (forward/backward repeated many times per epoch)
15	A, B, C, D, E	All statements are correct
16	A, B, C, D	E is false (sigmoid causes vanishing, doesn't prevent it)
17	A - True	Direct description from the content
18	A - True	Direct description from gradient descent section
19	A - True	Direct description from BGD section
20	A - True	Direct description from vanishing/exploding gradient section
21	forward propagation, error/loss, loss function, backpropagation, parameter adjustment	Neural network learning process
22	entire, one random, small batch, commonly	BGD vs SGD vs MBGD comparison
23	$f'(g(x)) \cdot g'(x)$, product, loss C, weight W_1	Chain rule application in backpropagation
24	0.25, 1/4, gradient vanishing, ReLU, LSTM	Sigmoid derivative and vanishing gradient solutions
25	A→2, B→3, C→1	Training=learn, Validation=adjust, Test=final evaluation
26	A→2, B→1, C→3	MSE=regression, Cross-entropy=classification, Hinge=SVM
27	A→3, B→1, C→2	BGD=slow/stable, SGD=fast/noisy, MBGD=balanced/common
28	A→2, B→1, C→3	Vanishing=small derivatives, Exploding=large weights, Slow=full dataset
29	A→2, B→1, C→3	Clipping=exploding, ReLU=vanishing (deep), LSTM=vanishing (sequences)
30	B→A→C→D	Forward → Compute loss → Backprop → Update parameters

Module 3.2.4: Optimizers & 3.2.5: Regularization — 30 Questions

Single Choice Questions (1-10)

Q1 A Huawei AI team is training a deep neural network for autonomous driving. The training is extremely slow with basic SGD, and the loss oscillates wildly near convergence. They need an optimizer that accelerates convergence while maintaining stability.

Which optimizer is MOST appropriate?

- A. Basic SGD without any modifications
- B. Momentum optimizer with inertia term
- C. AdaGrad with constant learning rate
- D. Step function optimizer

Q2 The Momentum optimizer update rule is:

$$\Delta w_{ji}^l(n) = -\eta \delta_i^{l+1} x_j^l(n) + \alpha \Delta w_{ji}^l(n-1)$$

What does the $\alpha \Delta w_{ji}^l(n-1)$ represent, and what is its effect?

- A. It is the current gradient; it increases the step size
- B. It is the momentum term from previous update; it gives inertia to help overshoot flat areas and local extrema
- C. It is the learning rate; it controls the step size
- D. It is the weight decay term; it prevents overfitting

Q3 AdaGrad uses a gradient accumulation variable r_t

that continuously increases. The update is: $\Delta w_t = -\frac{\eta}{\epsilon + \sqrt{r_t}} g_t$

What happens to the effective learning rate as training progresses?

- A. It increases exponentially
- B. It decreases because r_t increases in the denominator
- C. It remains constant
- D. It oscillates randomly

Q4 RMSProp improves upon AdaGrad by introducing an attenuation coefficient β : $r_t = \beta r_{t-1} + (1 - \beta) g_t^2$

What problem does this solve?

- A. RMSProp makes the learning rate increase over time
- B. RMSProp prevents r_t from growing too large, avoiding the "learning becomes very slow at the end" problem of AdaGrad
- C. RMSProp eliminates the need for a learning rate
- D. RMSProp only works for convex functions

Q5 Adam maintains two variables: m_t (moving average of gradient) and v_t (moving average of squared gradient). The bias-corrected estimates are: $\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$, $\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$ Why are bias corrections needed?

- A. Because m_t and v_t are initialized to zero and are biased towards zero in early iterations
- B. Because the learning rate needs to be increased
- C. Because gradients are always negative
- D. Because bias correction eliminates the need for backpropagation

Q6 The default settings for Adam are: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 0.001$

What is the PRIMARY advantage of Adam over other optimizers?

- A. It requires no hyperparameters at all
- B. It converges quickly with minimal manual tuning; only η may need reduction at saturation
- C. It guarantees convergence to the global minimum
- D. It works only for classification tasks

Q7 A model achieves 99% accuracy on training data but only 65% on test data. The model has 20 hidden layers with 1024 neurons each, trained on 5,000 images.

What is the root cause?

- A. Underfitting due to insufficient model capacity
- B. Overfitting due to high model complexity and insufficient training data
- C. The learning rate is too small
- D. The optimizer is not converging

Q8 L1 regularization: $\tilde{J}(w) = J(w) + \alpha|w|_1$

L2 regularization: $\tilde{J}(w) = J(w) + \alpha|w|_2$

A feature selection task needs to eliminate irrelevant features by driving their weights to exactly zero. Which regularization should be used?

- A. L1 regularization — drives weights to exactly zero
- B. L2 regularization — reduces weights but rarely to zero
- C. No regularization — let the model decide
- D. Both L1 and L2 are equally effective for feature selection

Q9 The early stopping diagram shows training loss continuously decreasing while validation loss decreases then increases.

At which point should training stop?

- A. When training loss reaches zero
- B. When validation loss is at its minimum, before it starts increasing
- C. After 1000 epochs regardless of validation loss
- D. When the model memorizes all training data

Q10 Dropout randomly discards neurons during training with probability p (e.g., 0.5 for hidden layers). During testing, all neurons are used. Why does this prevent overfitting?

- A. It reduces the number of parameters permanently
- B. It trains an ensemble of sub-networks, combining their results implicitly
- C. It increases the learning rate
- D. It eliminates the need for backpropagation

Multiple Choice Questions (11-16)  (Select ALL that apply)

Q11 Which statements about optimizers are CORRECT? (Select all that apply)

- A. SGD, MBGD, and Momentum update all parameters with the same learning rate
- B. AdaGrad sets different learning rates for different parameters based on gradient history
- C. RMSProp introduces an attenuation coefficient to prevent r_t from growing too large
- D. Adam combines ideas from AdaGrad and Momentum with bias correction
- E. All optimizers guarantee convergence to the global minimum

Q12 Which statements about the Momentum optimizer are CORRECT? (Select all that apply)

- A. It enhances stability of gradient correction direction and reduces mutations
- B. In stable gradient areas, it accelerates convergence by building up speed
- C. It helps roll over narrow local extrema
- D. Both learning rate η and momentum α need manual tuning
- E. It eliminates the need for a learning rate

Q13 Which statements about AdaGrad are CORRECT? (Select all that apply)

- A. r_t accumulates squared gradients and continuously increases
- B. The effective learning rate decreases as r_t increases
- C. The learning rate is automatically updated without manual intervention
- D. The denominator increases continuously, making learning very slow at the end
- E. It is the best optimizer for all deep learning tasks

Q14 Which are VALID methods to prevent overfitting? (Select all that apply)

- A. Obtain more data and perform data augmentation
- B. Reduce network layers and neurons to limit fitting capability
- C. Use Dropout to randomly discard neurons during training
- D. Apply L1/L2 regularization to restrict weight increase
- E. Use early stopping when validation loss starts increasing

Q15 Which statements about L1 and L2 regularization are CORRECT? (Select all that apply)

- A. L1 uses absolute value penalty ($\alpha|w|_1$); L2 uses squared penalty ($\frac{1}{2}\alpha|w|_2^2$)
- B. L1 can drive weights to exactly zero, performing feature selection
- C. L2 reduces weights by a factor $(1-\epsilon\alpha)$ but rarely to exactly zero
- D. L1 optimization: $w_{t+1} = (w - \epsilon\alpha \cdot \text{sign}(w)) - \epsilon\nabla J(w)$
- E. L2 is always better than L1 for all tasks

Q16 Which statements about Dropout are CORRECT? (Select all that apply)

- A. Dropout randomly discards neurons during training with probability p
- B. Parameters of discarded neurons are not updated in that iteration
- C. It is an integration method combining sub-network results
- D. Input layer dropout probability is typically 0.8; hidden layer is 0.5
- E. Dropout is less effective when training data is insufficient

True or False Questions (17-20)

Q17 The Momentum optimizer adds a momentum term $\alpha\Delta w(n-1)$ where $0 \leq \alpha < 1$, giving inertia to the weight update like a ball rolling down an error surface.

- A. True
- B. False

Q18 RMSProp is especially effective for RNNs because it handles unstable targets by preventing the gradient accumulation variable from growing too large.

- A. True
- B. False

Q19 In L1 regularization, if $|w_i^*| < \frac{\alpha}{\lambda_i}$, the parameter value is reduced to exactly 0. In L2 regularization, the parameter is reduced to a value close to 0 but not exactly 0.

- A. True
- B. False

Q20 Early stopping monitors the validation set loss during training. If the validation loss increases while training loss decreases, this indicates overfitting, and training should be stopped before the test error starts to increase.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 Common reasons for improving gradient descent algorithms include: _____ algorithm convergence, preventing or overshooting _____, and simplifying manual parameter setting especially the _____.

Q22 The Momentum optimizer weight correction rule is $\Delta w_{ji}^l(n) = -\eta \delta_i^{l+1} x_j^l(n) + \alpha \Delta w_{ji}^l(n-1)$ the term $\alpha \Delta w_{ji}^l(n-1)$ is called the _____, where α is the _____ ($0 \leq \alpha < 1$). This gives _____ to the weight update, helping the ball roll over _____.

Q23 AdaGrad accumulates squared gradients in $r_t = r_{t-1} + g_t^2$. The update is $\Delta w_t = -\frac{\eta}{\epsilon + \sqrt{r_t}} g_t$. As r_t _____, the learning rate _____. RMSProp fixes this by using $r_t = \beta r_{t-1} + (1 - \beta) g_t^2$ with _____ coefficient β . Adam further improves by tracking _____ m_t and _____ v_t with _____.

Q24 Overfitting occurs when a model performs well on _____ data but poorly on _____ data. Root causes include too many _____, too many _____, too much _____, and too little _____. Solutions include: more data, proper model size, _____, _____ (L1/L2), early _____, and _____.

Drag and Drop / Matching Questions (25-30)

Q25 Match each optimizer to its key feature:

Optimizer	Key Feature
A. Momentum	1. Different learning rates per parameter based on gradient history
B. AdaGrad	2. Moving averages of gradient and squared gradient with bias correction
C. RMSProp	3. Inertia term to accelerate through flat areas
D. Adam	4. Attenuation coefficient to prevent gradient accumulation explosion

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each optimizer to the problem it primarily addresses:

Optimizer	Problem Solved
A. Momentum	1. Same learning rate for all parameters
B. AdaGrad	2. Gradient oscillation near extremum
C. RMSProp	3. Learning becomes very slow at the end
D. Adam	4. Need for extensive manual hyperparameter tuning

Write your answer as: A→?, B→?, C→?, D→?

Q27 Match each regularization to its effect on weights:

Regularization	Effect
A. L1 regularization	1. Weights reduced by factor $(1 - \epsilon\alpha)$, rarely to zero
B. L2 regularization	2. Some weights driven exactly to zero (feature selection)
C. Dropout	3. Randomly sets neuron outputs to zero during training
D. Early stopping	4. Stops training before overfitting begins

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each overfitting cause to its solution:

Cause Solution

- | | |
|-------------------------------|--------------------------|
| A. High model complexity | 1. Dropout |
| B. Insufficient training data | 2. Reduce layers/neurons |
| C. Too much noise | 3. Data augmentation |
| D. Unstable weight updates | 4. L1/L2 regularization |

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each Adam component to its description:

Component Description

- | | |
|----------|---|
| A. m_t | 1. Moving average of squared gradient (second moment) |
| B. v_t | 2. Moving average of gradient (first moment) |
| C. m^t | 3. Bias-corrected second moment estimate |
| D. v^t | 4. Bias-corrected first moment estimate |

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each training scenario to the best solution:

Scenario Solution

- | | |
|---|--------------------------------|
| A. Training loss ↓, validation loss ↑ after epoch 50 | 1. Early stopping |
| B. Model has 50 layers, 10,000 neurons, trained on 1,000 images | 2. Dropout + L2 regularization |
| C. Gradients oscillate wildly near convergence | 3. Momentum optimizer |
| D. Some features are clearly irrelevant | 4. L1 regularization |
| E. Learning rate too large, overshooting minimum | 5. AdaGrad or Adam |

Write your answer as: A→?, B→?, C→?, D→?, E→?

Q	Answer	Explanation	
1	B	Momentum adds inertia, accelerates convergence, reduces oscillation	
2	B	$\alpha \Delta w_{(n-1)}$ is momentum term; gives inertia to overshoot flat areas/local extrema	
3	B	As r_t increases, denominator increases $\rightarrow$ effective learning rate decreases	
4	B	RMSProp's attenuation prevents r_t explosion, fixing AdaGrad's slow-end problem	
5	A	m_t, v_t initialized to zero $\rightarrow$ biased low; bias correction fixes this	
6	B	Adam converges quickly; defaults work well; only η may need tuning	
7	B	99% train vs 65% test = classic overfitting; complex model + small data	
8	A	L1 drives weights to exactly zero $\rightarrow$ feature selection; L2 rarely reaches zero	
9	B	Stop at minimum validation loss before overfitting starts	
10	B	Dropout trains ensemble of sub-networks implicitly	
11	A, B, C, D	E is false — no optimizer guarantees global minimum for non-convex functions	
12	A, B, C, D	E is false — Momentum still needs learning rate η	
13	A, B, C, D	E is false — AdaGrad has slow-end problem, not best for all tasks	
14	A, B, C, D, E	All are valid overfitting prevention methods	
15	A, B, C, D	E is false — L1 vs L2 depends on task (feature selection vs weight shrinkage)	
16	A, B, C, D, E	All statements about Dropout are correct	
17	A - True	Direct description from Momentum section	
18	A - True	RMSProp is especially effective for RNNs with unstable targets	
19	A - True	L1 drives to zero when η	$w_i^* < \alpha / \lambda_i$; L2 reduces but not to zero
20	A - True	Direct description from early stopping section	
21	accelerating, local extrema, learning rate	Reasons for improving optimizers	
22	momentum term, momentum coefficient, inertia, narrow local extrema	Momentum optimizer components	
23	increases, decreases, attenuation, first moment, second moment, bias correction	AdaGrad $\rightarrow$ RMSProp $\rightarrow$ Adam evolution	
24	training, test/new, feature dimensions, parameters, noise, training data, Dropout, regularization, stopping, ensemble methods	Overfitting causes and solutions	
25	A $\rightarrow$ 3, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 2	Momentum=inertia, AdaGrad=per-param LR, RMSProp=attenuation, Adam=bias correction	
26	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	Momentum=oscillation, AdaGrad=same LR, RMSProp=slow end, Adam>manual tuning	
27	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	L1=zero weights, L2=factor reduction, Dropout=random zero, Early stopping=stop before overfit	
28	A $\rightarrow$ 2, B $\rightarrow$ 3, C $\rightarrow$ 1, D $\rightarrow$ 4	Complexity $\rightarrow$ reduce size, Insufficient data $\rightarrow$ augmentation, Noise $\rightarrow$ dropout, Unstable $\rightarrow$ regularization	
29	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 3	m_t =first moment, v_t =second moment, $\hat{m}_t$ =corrected first, $\hat{v}_t$ =corrected second	
30	A $\rightarrow$ 1, B $\rightarrow$ 2, C $\rightarrow$ 3, D $\rightarrow$ 4, E $\rightarrow$ 5	Validation loss $\uparrow$ =early stopping, Complex model+small data=dropout+L2, Gradient oscillation=momentum, Irrelevant features=L1, Large LR=AdaGrad/Adam	

Module 3.3: Convolutional Neural Network — 30 Questions

Single Choice Questions (1-10)

Q1 CNN was inspired by Hubel and Wiesel's research in the 1960s on cats' cortex neurons. What was their key discovery that led to CNN architecture?

- A. Neurons process global image information simultaneously
- B. Neurons respond to local stimuli and have hierarchical feature processing
- C. Neurons use backpropagation for learning
- D. Neurons require fully-connected layers for vision

Q2 A CNN for image classification processes a 224×224 RGB image through multiple layers before outputting class probabilities.

What is the CORRECT sequence of layers?

- A. Input → Convolution → ReLU → Pooling → Fully-connected → Output
- B. Input → Pooling → Convolution → Fully-connected → ReLU → Output
- C. Input → Fully-connected → Convolution → Pooling → Output
- D. Input → Convolution → Pooling → Fully-connected → ReLU → Output

Q3 The concept of "local receptive field" in CNNs is based on what biological principle?

- A. Human perception of the outside world is from local to global
- B. Neurons process the entire image at once
- C. All pixels in an image are equally important regardless of distance
- D. Global features are extracted before local features

Q4 A CNN uses 3 filters (kernels) to scan a 256×256 image. Each filter is 3×3.

How many parameters does each filter have, and why is this efficient compared to fully-connected layers?

- A. 9 parameters per filter; efficiency comes from weight sharing across all spatial locations
- B. 256 parameters per filter; each pixel has its own weight
- C. 196,608 parameters per filter; fully-connected would need fewer
- D. 0 parameters; filters are randomly generated

Q5 An image is convolved with a 3×3 filter. The top-left 3×3 region of the image has values:

1 1 1

0 1 1

0 0 1

The filter weights are:

1 0 1

0 1 0

1 0 1

What is the convolved feature value at this position?

- A. 4
- B. 5
- C. 6
- D. 7

Q6 Max pooling with a 2x2 window and stride 2 is applied to:

25 8 9 48

56 1 5 6

35 255 0 4

12 2 32 2

What is the output?

A.

25 9

56 5

B.

56 48

255 32

C.

25 48

35 4

D.

56 6

255 32

Q7 Which statement about pooling layers is CORRECT?

A. Pooling increases the spatial dimensions of feature maps

B. Max pooling provides translational invariance within the pool window

C. Pooling requires learnable parameters

D. Average pooling always outperforms max pooling

Q8 After convolution and pooling, the feature maps are "straightened" (flattened) before the fully-connected layer.

What is the purpose of this flattening operation?

A. To reduce the number of layers in the network

B. To convert 2D feature maps into a 1D vector for classification

C. To increase the spatial dimensions

D. To eliminate the need for activation functions

Q9 Why can CNNs avoid complex image pre-processing and directly use original images?

A. Because CNNs have fewer parameters than fully-connected networks

B. Because convolution and pooling automatically extract hierarchical features from raw pixels

C. Because CNNs do not need activation functions

D. Because CNNs always use grayscale images

Q10 In a CNN with multiple convolutional layers, what happens to the features as we go deeper?

A. Features become simpler (e.g., from edges to pixels)

B. Features become more complex (e.g., from edges to object parts to objects)

C. Features remain the same complexity at all layers

D. Deep layers extract only color information

Multiple Choice Questions (11-16) ⚠ (Select ALL that apply)

Q11 Which are the MAIN concepts that make CNNs effective for image processing? (Select all that apply)

- A. Local receptive field
- B. Parameter sharing
- C. Fully-connected layers only
- D. Hierarchical feature extraction
- E. Global connections in early layers

Q12

Which statements about convolutional layers are CORRECT? (Select all that apply)

- A. Each convolutional unit's parameters are learned through backpropagation
- B. The first convolutional layer extracts low-level features like edges and lines
- C. Deeper layers extract more complex features based on lower-level features
- D. After convolution, results are biased and activated before next layer
- E. Convolutional layers always increase the spatial dimensions

Q13 Which are VALID functions of the pooling layer? (Select all that apply)

- A. Invariance: max pooling ensures output is invariant to small translations
- B. Dimension reduction: reduces input size for next layer
- C. Fixed-length output: can produce fixed-length from variable-length inputs
- D. Scale increase: extracts features from larger scales
- E. Overfitting prevention: simplifies network and reduces fitting precision

Q14 Compared to fully-connected networks for image classification, CNNs have which advantages? (Select all that apply)

- A. Fewer parameters due to weight sharing
- B. Preservation of spatial structure through local connections
- C. Ability to handle variable input sizes with proper pooling
- D. No need for any activation functions
- E. Direct use of raw images without complex pre-processing

Q15 A CNN uses multiple filters at each convolutional layer. Which statements are CORRECT? (Select all that apply)

- A. Each filter scans the entire image with the same parameters
- B. Different filters extract different features (e.g., edges, textures)
- C. The number of output feature maps equals the number of filters
- D. Filter parameters change during scanning
- E. Filter size is typically smaller than the input image

Q16 Which statements describe the complete CNN pipeline correctly? (Select all that apply)

- A. Input image → Convolution + nonlinearity → Pooling → (repeat) → Flatten → Fully-connected → Softmax output
- B. Convolutional layers extract local features; pooling reduces dimensions
- C. Fully-connected layer integrates all local features into global features
- D. Softmax outputs probability distribution over classes
- E. ReLU is commonly used as activation after convolution

True or False Questions (17-20)

Q17 CNN is a feedforward neural network that excels at image processing, including convolutional layers, pooling layers, and fully-connected layers.

- A. True
- B. False

Q18 Parameter sharing in CNNs means that when a convolution kernel scans an entire image, its parameters (weights) remain constant across all spatial locations.

- A. True
- B. False

Q19 Max pooling with a 2×2 window outputs the maximum value within each 2×2 region, providing translational invariance and reducing the spatial dimensions by half in each direction.

- A. True
- B. False

Q20 The fully-connected layer in a CNN is essentially a classifier that takes flattened features from convolutional/pooling layers and outputs class scores using Softmax.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 A CNN is a _____ neural network that excels at _____ processing. Its main components are the _____ layer, _____ layer, and _____ layer.

Q22 The two core concepts that make CNNs efficient are: (1) _____ — each neuron only needs to know _____ information, and (2) _____ — the same _____ (kernel) scans the entire image with _____ parameters.

Q23 In the convolution operation, a _____ matrix (kernel) slides over the input image, performing element-wise _____ and _____. The first convolutional layer extracts _____ features like _____, _____, and _____. Deeper layers extract _____ features based on these.

Q24

Pooling combines _____ units to reduce dimensions. _____ pooling selects the _____ value; _____ pooling selects the _____ value. Pooling provides _____ (invariance to small translations), reduces _____ size, obtains _____-length outputs, and helps _____ overfitting.

Drag and Drop / Matching Questions (25-30)

Q25) Match each CNN layer to its primary function:

Layer Function

- | | |
|--------------------|---|
| A. Convolutional | 1. Reduce spatial dimensions and provide invariance |
| B. ReLU | 2. Extract local features using learnable filters |
| C. Pooling | 3. Introduce nonlinearity |
| D. Fully-connected | 4. Classify integrated global features |

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match feature complexity to layer depth:**Feature Layer Depth**

- | | |
|-----------------------------------|------------------------------|
| A. Edges, lines, angles | 1. Deep layers (near output) |
| B. Object parts (eyes, wheels) | 2. Middle layers |
| C. Complete objects (faces, cars) | 3. Early layers (near input) |

Write your answer as: A→?, B→?, C→?**Q27 Match pooling type to its property:****Pooling Type Property**

- | | |
|------------------------------|--|
| A. Max pooling | 1. Outputs average value, smoother features |
| B. Average pooling | 2. Outputs maximum value, preserves strongest activation |
| C. 2×2 pooling with stride 2 | 3. Reduces spatial dimensions by half |

Write your answer as: A→?, B→?, C→?**Q28 Match CNN concept to its biological inspiration:****CNN Concept Biological Inspiration**

- | | |
|----------------------------|--|
| A. Local receptive field | 1. Visual cortex neurons respond to local stimuli |
| B. Hierarchical processing | 2. Simple cells → complex cells → hypercomplex cells |
| C. Parameter sharing | 3. Same type of neuron repeated across visual field |

Write your answer as: A→?, B→?, C→?**Q29 Match operation to its mathematical description:****Operation Formula**

- | | |
|--------------------|---|
| A. Convolution | 1. $\max(0, x)$ |
| B. ReLU activation | 2. $\frac{e^{z_j}}{\sum_k e^{z_k}}$ |
| C. Softmax output | 3. $\sum_{i,j} I_{i,j} \cdot K_{i,j}$ |
| D. Max pooling | 4. $\max_{i,j \in \text{window}} F_{i,j}$ |

Write your answer as: A→?, B→?, C→?, D→?**Q30 Match each scenario to the appropriate CNN design choice:****Scenario Design Choice**

- | | |
|---|---|
| A. Need to detect small objects in large images | 1. More convolutional layers with small filters |
| B. Need to reduce computation and prevent overfitting | 2. Add pooling layers |
| C. Need to classify 1000 different object categories | 3. Use multiple filters per layer |
| D. Need to handle color images | 4. 3-channel input (RGB) |
| E. Need to preserve exact spatial locations | 5. Use average pooling instead of max pooling |

Write your answer as: A→?, B→?, C→?, D→?, E→?

Q	Answer	Explanation
1	B	Hubel & Wiesel found neurons respond to local stimuli with hierarchical processing
2	A	Standard CNN: Conv → ReLU → Pool → ... → FC → Output
3	A	Local receptive field: humans perceive from local to global
4	A	3×3 filter = 9 parameters; shared across all locations = efficient
5	A	$(1 \times 1 + 1 \times 0 + 1 \times 1) + (0 \times 0 + 1 \times 1 + 1 \times 0) + (0 \times 1 + 0 \times 0 + 1 \times 1) = 1 + 0 + 1 + 0 + 1 + 0 + 1 = 4$
6	B	Top-left: $\max(25, 8, 56, 1) = 56$; Top-right: $\max(9, 48, 5, 6) = 48$; Bottom-left: $\max(35, 255, 12, 2) = 255$; Bottom-right: $\max(0, 4, 32, 2) = 32$
7	B	Max pooling provides translational invariance; reduces dimensions; no parameters
8	B	Flattening converts 2D feature maps to 1D vector for FC classification
9	B	CNNs automatically extract hierarchical features from raw images
10	B	Deep layers extract complex features (object parts → objects) from simple ones
11	A, B, D	C is false (FC only is not CNN); E is false (local not global connections)
12	A, B, C, D	E is false (convolution typically preserves or reduces dimensions)
13	A, B, C, D, E	All are valid pooling functions from the content
14	A, B, C, E	D is false (CNNs do use activation functions)
15	A, B, C, E	D is false (filter parameters are shared, don't change during scanning)
16	A, B, C, D, E	All describe the CNN pipeline correctly
17	A - True	Direct definition from CNN introduction
18	A - True	Direct definition of parameter sharing
19	A - True	Max pooling outputs max, provides invariance, halves dimensions with stride 2
20	A - True	Direct description of fully-connected layer function
21	feedforward, image, convolutional, pooling, fully-connected	CNN components
22	local receptive field, local, parameter sharing, filter, same/constant	Core CNN concepts
23	filter/kernel, multiplication, summation, low-level, edges, lines, angles, complex	Convolution operation and feature hierarchy
24	nearby, Max, maximum, Average, average, invariance, input, fixed, prevent	Pooling properties
25	A→2, B→3, C→1, D→4	Conv=extract features, ReLU=nonlinearity, Pool=reduce dimensions, FC=classify
26	A→3, B→2, C→1	Early=edges, Middle=parts, Deep=objects
27	A→2, B→1, C→3	Max=max value, Average=average value, 2×2 stride 2=halves dimensions
28	A→1, B→2, C→3	Local receptive field=local stimuli, Hierarchy=cell hierarchy, Sharing=neuron repetition
29	A→3, B→1, C→2, D→4	Convolution=sum of products, ReLU= $\max(0, x)$, Softmax=probabilities, Max pooling=max in window
30	A→1, B→2, C→3, D→4, E→5	Small objects=more layers small filters, Reduce computation=pooling, 1000 classes=multiple filters, Color=RGB input, Preserve location=average pooling

Module 3.4: RNN, LSTM & Seq2Seq — 40 Questions

Single Choice Questions (1-12)

Q1 An RNN processes a sequence of words: "I love machine learning". At time step $t=3$ (word "machine"), the hidden state S_3 depends on:

- A. Only the current input X_3
- B. Only the previous hidden state S_2
- C. Both current input X_3 AND previous hidden state S_2
- D. All previous inputs X_1, X_2, X_3 but not previous hidden states

Q2 A standard Feedforward Neural Network (FNN) receives a fixed-size input vector and produces a fixed-size output. An RNN differs fundamentally because:

- A. RNN uses more layers than FNN
- B. RNN maintains a hidden state that acts as memory across time steps, allowing it to process sequences of variable length
- C. RNN always produces larger outputs
- D. RNN does not use activation functions

Q3 Back-Propagation Through Time (BPTT) computes gradients by unrolling the RNN across time steps. For a sequence of 100 words, the gradient of the loss at time step 100 with respect to the weight W involves multiplying many Jacobian matrices.

What is the PRIMARY risk?

- A. The gradient becomes exactly zero immediately
- B. Vanishing or exploding gradients due to repeated multiplications across time steps
- C. The gradient always increases linearly
- D. BPTT is impossible for sequences longer than 50 time steps

Q4 In a standard RNN, a hint at the beginning of a long sentence may be forgotten by the end because:

- A. The network has too many parameters
- B. Long-term information attenuates as it passes through many time steps
- C. The learning rate is too large
- D. The network uses ReLU activation

Q5 LSTM was designed to solve the long-term memory problem of standard RNNs. The key innovation is:

- A. Using more hidden layers
- B. Introducing a cell state C_t and gating mechanisms (forget, input, output gates) to control information flow
- C. Eliminating the hidden state entirely
- D. Using only feedforward connections

Q6 The forget gate in LSTM computes: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$

What does $f_t = 0.95$ mean for a particular element in the cell state?

- A. 95% of the information from the previous cell state C_{t-1} will be retained
- B. 95% of the information will be discarded
- C. The input gate will allow 95% new information
- D. The output gate will release 95% of the information

Q7 The LSTM cell state update is: $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$

What does this equation achieve?

- A. It completely replaces old information with new information
- B. It selectively forgets old information (via f_t) and adds new candidate information (via i_t and $\tilde{C}_t$)
- C. It always keeps all old information and ignores new information
- D. It randomly selects information to keep

Q8 The LSTM output gate computes: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$, then $h_t = o_t \odot \tanh(C_t)$

What is the purpose of applying tanh to C_t before the output gate?

- A. To increase the cell state values
- B. To normalize C_t to range $(-1, 1)$ before selective output
- C. To eliminate negative values
- D. To make the gradient always equal to 1

Q9 In machine translation, the encoder-decoder architecture is needed because:

- A. Both input and output sequences have variable lengths
- B. The input is always longer than the output
- C. Fixed-size vectors cannot represent any information
- D. RNNs cannot be used for translation

Q10 A major challenge in Seq2Seq is that the context vector c has fixed length. For a very long input sentence (e.g., 500 words):

- A. The encoder becomes faster
- B. Information is compressed and lost, especially for early words
- C. The decoder can handle it without any issues
- D. The model automatically splits the sentence

Q11 Which RNN type is used for sentiment classification (input: a sentence, output: positive/negative)?

- A. One-to-many
- B. Many-to-one
- C. Many-to-many (same length)
- D. Many-to-many (different length)

Q12 During inference (translation), the decoder in Seq2Seq generates words one by one. The input to the decoder at time step t' is:

- A. The original encoder input
- B. The previous decoder output $y_{t'-1}$ and the context variable c
- C. Only the context variable c
- D. Random noise

Multiple Choice Questions (13-20)  (Select ALL that apply)

Which scenarios are suitable for RNNs? (Select all that apply)

- A. Video analysis (sequence of frames)
- B. Audio processing (sequence of clips)
- C. Sentence analysis (sequence of words)
- D. Image classification (single static image)
- E. Time series prediction

Q14 Which statements about RNN architecture are CORRECT? (Select all that apply)

- A. $S_t = \sigma(UX_t + WS_{t-1})$ shows the hidden state depends on current input and previous state
- B. $O_t = \text{softmax}(S_t V)$ produces output at each time step
- C. The same weight matrices U, W, V are shared across all time steps
- D. RNNs are restricted by space boundaries like FNNs
- E. RNNs can extend to arbitrary time sequences

Q15 Which statements about BPTT are CORRECT? (Select all that apply)

- A. BPTT is conventional backpropagation extended through time
- B. At each time step, two errors are combined: current error and error from next time step
- C. The total gradient of weight W is accumulated across all time steps
- D. Longer sequences always make training faster
- E. Vanishing/exploding gradients are more likely with longer sequences

Q16 Which statements about LSTM gates are CORRECT? (Select all that apply)

- A. Forget gate: decides what information to discard from cell state
- B. Input gate: decides what new information to store in cell state
- C. Candidate value $\tilde{C}_t$: created by tanh layer, represents new candidate information
- D. Output gate: decides what part of cell state to output as h_t
- E. All gates use ReLU activation

Q17 Which statements about Seq2Seq are CORRECT? (Select all that apply)

- A. Encoder converts variable-length input to fixed-shape context variable c
- B. Decoder maps fixed-shape c to variable-length output
- C. $c = q(h_1, \dots, h_T)$ combines all encoder hidden states
- D. Decoder uses previous output y_{t-1} and context c as inputs
- E. $\langle \text{eos} \rangle \langle \text{bos} \rangle$

Q18 Which are VALID challenges of Seq2Seq? (Select all that apply)

- A. Single fixed-length vector c may lose information for long sequences
- B. RNN-based Seq2Seq does not support parallel computing
- C. Earlier information tends to be lost in long sequences
- D. During inference, words must be generated sequentially (slow)
- E. CNN-based Seq2Seq has fewer parameters than RNN-based

Q19 Match the following tasks to RNN types. Which are CORRECT? (Select all that apply)

- A. Image captioning (image $\rightarrow$ sentence): One-to-many
- B. Sentiment analysis (sentence $\rightarrow$ rating): Many-to-one
- C. Machine translation (sentence $\rightarrow$ sentence): Many-to-many (different length)
- D. Video frame labeling (video $\rightarrow$ labels per frame): Many-to-many (same length)
- E. Standard classification (image $\rightarrow$ label): One-to-one

Q20 Which statements correctly compare LSTM and standard RNN? (Select all that apply)

- A. LSTM has a cell state in addition to hidden state
- B. LSTM uses gating mechanisms to control information flow
- C. LSTM is better at capturing long-term dependencies
- D. Standard RNN has simpler architecture but worse long-term memory
- E. LSTM eliminates vanishing gradients completely

True or False Questions (21-24)

Q21 RNNs can keep a context state and store, learn, and express related information in context windows of any length, unlike FNNs which are restricted by space boundaries.

- A. True
- B. False

Q22 In BPTT, the total gradient of weight W is the accumulation of gradients across all time steps, and longer sequences increase the likelihood of vanishing or exploding gradients.

- A. True
- B. False

Q23 The LSTM forget gate outputs values in $[0, 1]$. A value of 1 means "completely retain this information," while 0 means "completely discard this information."

- A. True
- B. False

Q24 In Seq2Seq, the encoder processes the entire input sequence first, producing a context vector c . Then the decoder generates the output sequence one word at a time, using c and its previous output.

- A. True
- B. False

Fill in the Blank Questions (25-28)

Q25 An RNN is a neural network that captures _____ information in _____ data through periodic connections. Unlike FNNs, it is not restricted by _____ boundaries but can extend _____ sequences.

Q26 The LSTM cell state update equation is $C_t = \text{_____} \odot C_{t-1} + \text{_____} \odot \tilde{C}_t$. The forget gate $f_t = \sigma(W_f \cdot [\text{_____, _____}] + b_f)$ outputs values in _____. The input gate i_t decides what _____ information to store. The output gate o_t decides what part of C_t to _____ as h_t .

Q27 In Seq2Seq, the _____ converts variable-length input to fixed-shape _____ variable c . The _____ maps c to variable-length output. The context $c = q(\text{_____, ...}, \text{_____})$ combines all encoder _____ states. During inference, the decoder is _____ because each word depends on the _____ output.

Q28 BPTT computes gradients by _____ the RNN across time steps. At each step, gradients flow _____ from output to input. The challenge is that gradients may _____ (become very small) or _____ (become very large) when multiplied across many time steps. LSTM mitigates this through _____ and _____ mechanisms.

Q29 Match each LSTM gate to its function:

Gate Function

- | | |
|----------------|---|
| A. Forget gate | 1. Decide what new information to store |
| B. Input gate | 2. Decide what to output from cell state |
| C. Output gate | 3. Decide what to discard from cell state |

Write your answer as: A→?, B→?, C→?

Q30 Match each LSTM component to its equation:

Component Equation

- | | |
|----------------------|--|
| A. Forget gate | 1. $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$ |
| B. Input gate | 2. $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ |
| C. Candidate value | 3. $\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$ |
| D. Cell state update | 4. $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$ |

Write your answer as: A→?, B→?, C→?, D→?

Q31 Match each RNN type to its typical task:

RNN Type Task

- | | |
|------------------------------------|----------------------------|
| A. One-to-one | 1. Image captioning |
| B. One-to-many | 2. Standard classification |
| C. Many-to-one | 3. Machine translation |
| D. Many-to-many (same length) | 4. Sentiment analysis |
| E. Many-to-many (different length) | 5. Video frame labeling |

Write your answer as: A→?, B→?, C→?, D→?, E→?

Q32 Match each Seq2Seq challenge to its underlying cause:

Challenge Cause

- | | |
|-------------------------------------|--|
| A. Information loss for long inputs | 1. Fixed-length context vector c |
| B. No parallel computing | 2. RNN sequential dependency |
| C. Slow inference | 3. Autoregressive word generation |
| D. Early information lost | 4. Information compression in long sequences |

Write your answer as: A→?, B→?, C→?, D→?

Q33 Match each problem to its characteristic:

Problem Characteristic

- | | |
|---------------------------------|---|
| A. Vanishing gradient | 1. Gradients become extremely large |
| B. Exploding gradient | 2. Long-term information fades away |
| C. Long-term memory attenuation | 3. Gradients become extremely small |
| D. Fixed context bottleneck | 4. Single vector represents entire sequence |

Write your answer as: A→?, B→?, C→?, D→?

Q34 Match each activation function to where it's used in LSTM:

Activation Gate/Operation

- | | |
|---------------------------|--|
| A. Sigmoid (σ) | 1. Candidate value generation |
| B. Tanh | 2. All gating mechanisms (forget, input, output) |
| C. Both σ and tanh | 3. Cell state normalization before output |

Write your answer as: A→?, B→?, C→?

Q35 Match each element to its description:

Element	Description
A. <eos>	1. Start of decoder generation
B. <bos>	2. End of encoder processing
C. Context variable c	3. Previous decoder output used as next input
D. y_{t-1}	4. Fixed-shape representation of entire input

Write your answer as: A→?, B→?, C→?, D→?

Q36 Match each LSTM state to its purpose:

State	Purpose
A. Cell state C_t	1. Short-term output for current time step
B. Hidden state h_t	2. Long-term memory carrying information across time
C. Candidate $\tilde{C}_t$	3. Proposed new information to add to memory

Write your answer as: A→?, B→?, C→?

Scenario-Based Questions (37-40)

Q37 An LSTM processes the sentence: "The cat, which was very hungry and had been wandering for hours, sat on the mat." The word "cat" appears at the beginning and "sat" at the end.

When processing "sat", the forget gate for the "cat" information has $f_t \approx 0.99$. What does this indicate, and why is it important?

- A. The LSTM has forgotten the subject; this is a failure mode
- B. The LSTM retains the subject information; this enables correct subject-verb agreement
- C. The input gate is blocking new information
- D. The output gate is preventing any output

Q38 A Seq2Seq model translates "我爱机器学习" (Chinese) to "I love machine learning" (English). The encoder LSTM processes all Chinese characters and produces context c . The decoder LSTM then generates English words.

If the input sentence is extended to 1000 characters, what problem arises and why?

- A. The decoder becomes faster because it has more context
- B. Information loss increases because c is fixed-length and must compress more information
- C. The encoder can handle it without any issues
- D. The model automatically increases c 's dimension

Q39 Two models are trained on a long text classification task (documents with 500+ words):

Model A: Standard RNN

Model B: LSTM

Model A achieves 60% accuracy while Model B achieves 85%. What explains this difference?

- A. LSTM has more parameters simply because it's larger
- B. LSTM's gating mechanisms preserve long-term dependencies better than standard RNN
- C. RNN is inherently better for long sequences
- D. The training data is biased towards LSTM

Q40 An encoder-decoder model is used for speech recognition: audio waveform → text transcript.

Which statement about the encoder and decoder is CORRECT?

- A. Encoder processes variable-length audio; decoder produces variable-length text
- B. Both encoder and decoder process fixed-length inputs/outputs
- C. Encoder produces variable-length output; decoder takes fixed-length input
- D. The encoder and decoder are identical architectures with identical roles

 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	C	$S_3 = \sigma(UX_3 + WS_2)$ depends on both X_3 and S_2
2	B	RNN's hidden state acts as memory across time steps
3	B	Repeated Jacobian multiplications cause vanishing/exploding
4	B	Long-term information attenuates through many time steps
5	B	LSTM's cell state + gates control information flow
6	A	$f_t = 0.95$ means 95% retention of C_{t-1} information
7	B	Selective forget (f_t) + selective add ($i_t, \tilde{C}_t$)
8	B	$\tanh$ normalizes C_t to $(-1, 1)$ before output selection
9	A	Both input and output sequences have variable lengths
10	B	Fixed c compresses long input, losing information
11	B	Sentence $\rightarrow$ single label = many-to-one
12	B	Decoder uses previous output y_{t-1} and context c
13	A, B, C, E	D is false (single image = FNN/CNN, not RNN)
14	A, B, C, E	D is false (RNNs are NOT restricted by space boundaries)
15	A, B, C, E	D is false (longer sequences make training harder, not faster)
16	A, B, C, D	E is false (gates use sigmoid, not ReLU)
17	A, B, C, D, E	All statements about Seq2Seq are correct
18	A, B, C, D	E is false (CNN-based has MORE parameters, not fewer)
19	A, B, C, D, E	All task-type matches are correct
20	A, B, C, D	E is false (LSTM mitigates but doesn't eliminate vanishing gradients)
21	A - True	Direct description of RNN capabilities
22	A - True	Direct description of BPTT and gradient problems
23	A - True	Direct description of forget gate behavior
24	A - True	Direct description of Seq2Seq inference
25	dynamic, sequential, space, time	RNN fundamentals
26	$f_t, i_t, h_{t-1}, x_t, \setminus [0,1], \text{new}, \text{output}$	LSTM gate equations
27	encoder, context, decoder, $h_1, h_{\setminus T}$, hidden, autoregressive, previous	Seq2Seq architecture
28	unrolling, backward, vanish, explode, cell state, gating	BPTT and LSTM solutions
29	A $\rightarrow$ 3, B $\rightarrow$ 1, C $\rightarrow$ 2	Forget=discard, Input=store, Output=output
30	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 4	Forget= f_t , Input= i_t , Candidate= $\tilde{C}_t$, Update= C_t equation
31	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 5, E $\rightarrow$ 3	One-to-one=classification, One-to-many=captioning, Many-to-one=sentiment, Many-to-many same=video labeling, Many-to-many different=translation
32	A $\rightarrow$ 1, B $\rightarrow$ 2, C $\rightarrow$ 3, D $\rightarrow$ 4	Info loss=fixed c , No parallel=RNN sequential, Slow inference=autoregressive, Early lost=compression
33	A $\rightarrow$ 3, B $\rightarrow$ 1, C $\rightarrow$ 2, D $\rightarrow$ 4	Vanishing=small gradients, Exploding=large gradients, Attenuation=long-term fade, Bottleneck=single vector
34	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3	Sigmoid=all gates, Tanh=candidate value, Both=cell state update (σ for gates, $\tanh$ for candidate)
35	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 3	$\langle \text{eos} \rangle$ =end encoder, $\langle \text{bos} \rangle$ =start decoder, c =fixed representation, y_{t-1} =previous output
36	A $\rightarrow$ 2, B $\rightarrow$ 1, C $\rightarrow$ 3	Cell state=long-term memory, Hidden state=short-term output, Candidate=proposed new info
37	B	High $f_t \approx 0.99$ retains subject, enabling correct agreement
38	B	Longer input $\rightarrow$ more compression of fixed $c \rightarrow$ more information loss
39	B	LSTM gates preserve long-term dependencies better
40	A	Encoder: variable audio $\rightarrow$ fixed c ; Decoder: fixed $c \rightarrow$ variable text

Module 3.5: Transformer Architecture — 40 Questions

Single Choice Questions (1-12)

Q1 The Transformer model was proposed in a 2017 paper titled "Attention Is All You Need" by Google. What was the revolutionary claim of this paper?

- A. RNNs with more layers always outperform CNNs
- B. Attention mechanisms alone, without CNN or RNN structures, can effectively process sequences
- C. Transformer requires 10x more parameters than RNN
- D. CNNs are superior to attention for all NLP tasks

Q2 The attention mechanism is inspired by human cognitive capabilities. When people observe a scene, they:

- A. Process every pixel with equal importance
- B. Focus on salient objects and ignore minor details
- C. Memorize the entire scene perfectly
- D. Only look at the center of the image

Q3 In the attention mechanism, a Query is matched against Keys to produce scores, which are then converted to weights via softmax and applied to Values.

In a library analogy where you search for "Marvel" books:

Books = Values

Book numbers = Keys

Your interest "Marvel" = Query

What does the attention output represent?

- A. The total number of books in the library
- B. A weighted combination of book contents based on relevance to "Marvel"
- C. Only the exact book titled "Marvel"
- D. A random selection of books

Q4 The attention computing process has three phases:

Calculate similarity scores between Query and Keys

Convert scores to probability distribution via softmax

Weighted summation of Values

What is the mathematical purpose of the softmax in Phase 2?

- A. To increase the magnitude of all scores
- B. To normalize scores to $[0, 1]$ so they sum to 1, creating a valid probability distribution
- C. To eliminate negative scores
- D. To randomly select which Values to use

Q5 Self-attention differs from standard attention because:

- A. It uses different computing process
- B. It computes attention between elements within the same sequence (source-to-source or target-to-target), not between different sequences
- C. It requires more parameters than standard attention
- D. It only works for images, not text

Q6 In the sentence "The Law will never be perfect, but its application should be just", self-attention is used to understand what "its" refers to.

If "its" is the Query and the sentence words are Keys/Values, which word would receive the highest attention weight?

- A. "perfect"
- B. "Law"
- C. "application"
- D. "just"

Q7 Which statement correctly describes a key advantage of Transformer over RNN?

- A. Transformer has more parameters than RNN
- B. Transformer supports parallel computing because attention steps don't depend on previous steps
- C. Transformer can only process short sequences
- D. Transformer requires sequential processing like RNN

Q8 The Transformer architecture consists of an Encoder and a Decoder. Which component is NOT part of the standard Transformer block?

- A. Multi-head Attention
- B. Feed Forward layer
- C. Add & Norm (residual connection + layer normalization)
- D. Convolutional layer

Q9 In the Decoder, there is "Masked Multi-head Attention" instead of regular Multi-head Attention.

What is the purpose of the mask?

- A. To prevent the decoder from attending to future positions during training
- B. To reduce the number of parameters
- C. To increase the learning rate
- D. To eliminate the need for positional encoding

Q10 Since Transformer has no recurrent or convolutional structure, it doesn't inherently know word order. How does Transformer handle sequence order?

- A. It ignores word order completely
- B. It adds positional encoding to the input embeddings
- C. It uses a separate RNN layer for ordering
- D. Words are automatically ordered alphabetically

Q11 Which of the following is NOT listed as an advantage of the attention mechanism?

- A. Fewer parameters compared to CNN and RNN
- B. Higher speed due to parallel processing capability
- C. Better results by capturing long-distance dependencies
- D. Guaranteed convergence to global minimum

Q12 In the Transformer architecture, how does the Decoder receive information from the Encoder?

- A. The Decoder directly copies Encoder outputs
- B. The Encoder outputs serve as Keys and Values for the Decoder's second multi-head attention layer
- C. The Decoder ignores the Encoder completely
- D. Through a recurrent connection between final layers

Multiple Choice Questions (13-20) ⚠ (Select ALL that apply)

Q13 Which statements about the attention mechanism are CORRECT? (Select all that apply)

- A. Attention can be used without the encoder-decoder framework
- B. The core idea is "weighted summation" based on relevance
- C. Attention requires sequential dependency between steps
- D. Attention focuses on important aspects while ignoring minor things
- E. Attention always increases model complexity

Q14 Which statements about self-attention are CORRECT? (Select all that apply)

- A. Self-attention operates between elements within the same sequence
- B. It uses the same computing process as standard attention
- C. It helps capture long-distance dependencies in text
- D. It is less dependent on external information
- E. It only works in the encoder, never in the decoder

Q15 Which components are present in the standard Transformer architecture? (Select all that apply)

- A. Embedding Layer
- B. Positional Encoding
- C. Multi-head Attention
- D. Feed Forward network
- E. Add & Norm (residual connections + layer normalization)

Q16 Which are VALID advantages of Transformer over RNN/CNN? (Select all that apply)

- A. Fewer parameters and less complexity
- B. Supports parallel processing
- C. Better at capturing long-distance dependencies
- D. Automatically captures relative associations at different positions
- E. Requires less training data

Q17 Which correctly describe the three phases of attention computing? (Select all that apply)

- A. Phase 1: Calculate similarity scores between Query and each Key
- B. Phase 2: Convert scores to probability distribution via softmax
- C. Phase 3: Use weights as matrix to perform weighted summation on Values
- D. Phase 1 requires knowing the final output first
- E. Phase 2 uses ReLU instead of softmax

Q18 Which statements correctly compare Seq2Seq (RNN-based) and Transformer? (Select all that apply)

- A. Seq2Seq uses a single fixed-length context vector; Transformer uses attention to access all encoder states
- B. Seq2Seq processes sequentially; Transformer processes in parallel
- C. Seq2Seq struggles with long sequences; Transformer handles long texts better
- D. Transformer completely rejects CNN and RNN structures
- E. Both use exactly the same attention mechanism

Q19 The content describes the evolution: RNN era (rote learning) → Attention model → Transformer → GPT/BERT.

Which statements about this evolution are CORRECT? (Select all that apply)

- A. RNN era is like rote learning (memorizing grammar)
- B. Attention model evolved into Transformer with excellent expression learning
- C. GPT and BERT are built on Transformer architecture
- D. Multi-task large-scale learning accumulated practical experience
- E. RNNs are still superior to Transformers for all tasks

Q20 Multi-head attention runs multiple attention mechanisms in parallel. Which statements are CORRECT? (Select all that apply)

- A. Each "head" can learn different types of relationships
- B. Outputs from all heads are concatenated and linearly transformed
- C. It allows the model to jointly attend to information from different representation subspaces
- D. It increases computational cost linearly with number of heads
- E. It is only used in the encoder, not the decoder

True or False Questions (21-24)

Q21 Transformer uses a whole new formula and rejects CNN and RNN structures, using only the Attention mechanism to capture relative associations at different positions.

- A. True
- B. False

Q22 Self-attention refers to the attention mechanism used between source elements or target elements, rather than between source and target. The computing process is the same, but the objects differ.

- A. True
- B. False

Q23 In the attention mechanism, the steps depend on the previous step (like RNN), which is why attention alone cannot support parallel processing.

- A. True
- B. False

Q24 Positional encoding is added to input embeddings in Transformer because, unlike RNNs, Transformers have no inherent notion of sequence order.

- A. True
- B. False

Fill in the Blank Questions (25-28)

Q25 The attention mechanism is inspired by human _____ capabilities. When observing a scene, humans focus on _____ objects and ignore _____ details. The core idea is " _____ " based on relevance scores.

Q26 The attention computing process has three phases: (1) Calculate _____ between _____ and each _____ to obtain scores; (2) Convert scores to _____ via _____; (3) Perform _____ on _____ using the weights to obtain the final Attention Value.

Q27 Transformer was proposed in the paper " _____ " (2017). It uses _____ mechanism to process sequences. Unlike RNN, it _____ CNN and RNN structures. Key components include: _____ Layer, _____ Encoding, _____-head Attention, _____ Forward, and Add & _____.

Q28 Self-attention computes attention between _____ within the same _____. For example, to understand what "its" refers to, "its" serves as the _____, while the sentence words serve as both _____ and _____. This helps address the - _____ problem in NLP.

Drag and Drop / Matching Questions (29-36)

Q29 Match each attention component to its role:

Component	Role
A. Query	1. The items being weighted and combined
B. Key	2. What we are looking for
C. Value	3. Identifiers used to calculate relevance scores

Write your answer as: A→?, B→?, C→?

Q30 Match each attention phase to its operation:

Phase	Operation
A. Phase 1	1. Weighted summation of Values
B. Phase 2	2. Calculate similarity scores
C. Phase 3	3. Softmax normalization

Write your answer as: A→?, B→?, C→?

Q31 Match each attention advantage to its explanation:

Advantage	Explanation
A. Fewer parameters	1. Captures long-distance dependencies better
B. Higher speed	2. Less complex than CNN/RNN
C. Better result	3. Supports parallel processing

Write your answer as: A→?, B→?, C→?

Q32 Match each architecture to its key characteristic:

Architecture	Characteristic
A. RNN	1. Uses only attention, no recurrence or convolution
B. CNN	2. Sequential processing with hidden state memory
C. Transformer	3. Local receptive fields with parameter sharing

Write your answer as: A→?, B→?, C→?

Q33 Match each Transformer component to its purpose:

Component	Purpose
A. Positional Encoding	1. Prevent attending to future positions
B. Masked Multi-head Attention	2. Provide sequence order information
C. Feed Forward	3. Transform representations non-linearly
D. Add & Norm	4. Stabilize training with residual connections

Write your answer as: A→?, B→?, C→?, D→?

Q34 Match each evolution stage to its learning analogy:

Stage	Analogy
A. RNN era	1. Grasping keywords in sentences
B. Attention model	2. Rote learning (memorizing grammar)
C. Transformer/GPT/BERT	3. Excellent expression learning capabilities

Write your answer as: A→?, B→?, C→?

Q35 Match each Seq2Seq problem to how Transformer solves it:

Problem

Solution

- A. Single fixed-length context vector
- B. Cannot parallelize
- C. Short memory / long-distance issues
- D. Poor at capturing relationships

- 1. Self-attention captures internal correlations
- 2. Attention accesses all encoder states directly
- 3. Parallel processing without sequential dependency
- 4. Direct connections between any two positions

Write your answer as: A→?, B→?, C→?, D→?

Q36 Match each attention type to its use case:

Attention Type Use Case

- A. Standard Attention (Encoder-Decoder)
- B. Self-Attention
- C. Masked Self-Attention

- 1. Understanding word relationships within a sentence
- 2. Machine translation between different languages
- 3. Preventing decoder from seeing future words

Write your answer as: A→?, B→?, C→?

Scenario-Based Questions (37-40)

Q37 A translation model translates "The cat sat on the mat because it was tired." The model must understand that "it" refers to "cat" not "mat."

How does self-attention help with this?

- A. It processes words sequentially from left to right
- B. It directly computes the relationship between "it" and all other words, giving high weight to "cat"
- C. It ignores pronouns entirely
- D. It uses a fixed-size context vector

Q38 Two teams train machine translation models:

Team A: RNN-based Seq2Seq

Team B: Transformer

Team B's model trains 10x faster on the same GPU hardware. What is the primary reason?

- A. Transformer has fewer parameters
- B. Transformer's attention mechanism supports parallel processing across sequence positions
- C. Transformer uses a smaller dataset
- D. Transformer doesn't need backpropagation

Q39 A legal document analysis system must process 10,000-word contracts and find relationships between clauses at the beginning and end.

Why would Transformer outperform RNN for this task?

- A. Transformer has more layers
- B. Transformer's self-attention provides direct connections between any two positions, regardless of distance
- C. Transformer uses less memory
- D. Transformer was trained on more legal documents

Q40 A company needs to build a system for:

Real-time speech recognition (audio → text)

Document summarization (long text → short text)

Sentiment analysis (text → positive/negative)

Which architecture is MOST suitable for all three tasks?

- A. CNN only
- B. RNN only
- C. Transformer-based models
- D. Perceptron only

 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	B	"Attention Is All You Need" claimed attention alone can replace RNN/CNN for sequences
2	B	Humans focus on salient objects, ignore minor details
3	B	Attention output = weighted combination based on relevance
4	B	Softmax normalizes to $[0,1]$ with sum=1, creating probability distribution
5	B	Self-attention = within same sequence (source-source or target-target)
6	B	"its" most relevant to "Law" (subject reference)
7	B	Transformer supports parallel computing; attention steps are independent
8	D	Transformer has NO convolutional layers
9	A	Mask prevents decoder from seeing future positions during training
10	B	Positional encoding provides order information
11	D	No optimizer/mechanism guarantees global minimum
12	B	Encoder outputs = Keys & Values for Decoder's cross-attention
13	A, B, D	C is false (attention steps are independent); E is false (attention reduces complexity)
14	A, B, C, D	E is false (self-attention is used in both encoder and decoder)
15	A, B, C, D, E	All are standard Transformer components
16	A, B, C, D	E is false (Transformer doesn't inherently require less data)
17	A, B, C	D is false (output not needed first); E is false (uses softmax, not ReLU)
18	A, B, C, D	E is false (Transformer uses self-attention; Seq2Seq uses different attention)
19	A, B, C, D	E is false (Transformers outperform RNNs in most NLP tasks)
20	A, B, C, D	E is false (multi-head used in both encoder and decoder)
21	A - True	Direct description from Transformer section
22	A - True	Direct description of self-attention
23	B - False	Attention steps do NOT depend on previous steps — supports parallelism
24	A - True	Positional encoding provides order that RNN gets from recurrence
25	cognitive, salient, minor, weighted summation	Attention inspiration and core idea
26	similarity, Query, Key, probability distribution, softmax, weighted summation, Values	Three phases of attention computing
27	Attention Is All You Need, self-attention, rejects, Embedding, Positional, Multi, Feed, Norm	Transformer paper and components
28	elements, sequence, Query, Keys, Values, long-distance dependency	Self-attention mechanism
29	A→2, B→3, C→1	Query=looking for, Key=identifiers, Value=items combined
30	A→2, B→3, C→1	Phase 1=scores, Phase 2=softmax, Phase 3=weighted sum
31	A→2, B→3, C→1	Fewer params=less complex, Speed=parallel, Better=long-distance
32	A→2, B→3, C→1	RNN=sequential, CNN=local fields, Transformer=attention only
33	A→2, B→1, C→3, D→4	Positional=order, Masked=future prevention, Feed Forward=non-linear, Add&Norm=residual
34	A→2, B→1, C→3	RNN=rote learning, Attention=keywords, Transformer=expression learning
35	A→2, B→3, C→4, D→1	Fixed vector→all states, No parallel→parallel, Short memory→direct connections, Poor relationships→internal correlations
36	A→2, B→1, C→3	Standard=translation, Self=word relationships, Masked=prevent future

37	B	Self-attention directly computes "it" ↔ "cat" relationship
38	B	Parallel processing = 10x speedup
39	B	Direct position-to-position connections regardless of distance
40	C	Transformer handles sequences for speech, text generation, and classification

Module 3.6: Basic Foundation Model Architecture — 30 Questions

Single Choice Questions (1-10)

Q1 What is the defining characteristic of foundation models like GPT, LLaMA, and GLM?

- A. They are trained on small datasets with limited compute
- B. They are trained with massive data and compute, achieving strong generation and generalization, and can adapt to new tasks via fine-tuning
- C. They can only perform one specific task
- D. They use only CNN architectures

Q2 How did Transformer enable the age of foundation models?

- A. By making models smaller and less complex
- B. Through its scalability and increased computing power, enabling billion-parameter models
- C. By eliminating the need for training data
- D. By using only RNN structures

Q3 GPT-1 uses the Transformer decoder framework. Which modification did GPT-1 make to the standard Transformer positional encoding?

- A. It used sine/cosine functions like the original Transformer
- B. It used random initialization similar to word embeddings, updated during training
- C. It eliminated positional encoding entirely
- D. It used a separate RNN layer for position information

Q4 GPT-1 expanded the Transformer decoder significantly. Which scaling change is CORRECT?

- A. Attention dimensions: 512 → 768; Attention heads: 8 → 12; Feed Forward hidden layers: 2048 → 3072; Total parameters: 117 million
- B. Attention dimensions: 768 → 512; Attention heads: 12 → 8; Total parameters: 1.17 billion
- C. Layers: 6 → 12; Total parameters: 1.7 million
- D. No changes from original Transformer

Q5 BERT (2018) is a bidirectional model trained on 3.3 billion words with 340 million parameters. What does "bidirectional" mean in BERT?

- A. It processes text from left to right only
- B. It analyzes the context of a complete sequence from both directions before making predictions
- C. It uses two separate models running in parallel
- D. It only processes text from right to left

Q6 Which statement correctly distinguishes BERT from GPT-1?

- A. BERT uses the Transformer encoder; GPT-1 uses the Transformer decoder
- B. BERT is unidirectional; GPT-1 is bidirectional
- C. BERT has fewer parameters than GPT-1
- D. Both use exactly the same architecture

Q7 In conventional learning, when a model updates weights for one scenario, it affects weights for other scenarios. What is this phenomenon called?

- A. Transfer learning
- B. Interference effect
- C. Gradient descent
- D. Attention mechanism

Q8 Mixture-of-Experts (MoE) combines multiple models ("experts") with a gating mechanism. What is the PRIMARY advantage of MoE?

- A. It reduces the total number of parameters
- B. It improves model capacity and efficiency, allowing effective pre-training with less compute than dense models
- C. It eliminates the need for fine-tuning
- D. It only works for image classification

Q9 Why is training a larger MoE model with fewer steps often better than training a smaller dense model with more steps?

- A. Because MoE models have fewer total parameters
- B. Because model size is a key factor for performance, and MoE enables scaling under limited compute budget
- C. Because dense models always overfit
- D. Because MoE doesn't need any training data

Q10 A foundation model trained on general text achieves excellent performance on question answering. It is then fine-tuned on medical texts to answer medical questions.

What does this demonstrate?

- A. The model can only do one task
- B. Quick adaptation to new tasks and data through fine-tuning
- C. The model forgets everything it learned before
- D. Fine-tuning always makes performance worse

Multiple Choice Questions (11-16)  (Select ALL that apply)

Q11 Which statements about foundation models are CORRECT? (Select all that apply)

- A. They are trained on massive amounts of data with significant compute resources
- B. They achieve excellent performance on single tasks
- C. They can quickly adapt to new tasks through fine-tuning
- D. Examples include GPT series, LLaMA, and GLM
- E. They are limited to only NLP tasks

Q12 Which statements about GPT-1 are CORRECT? (Select all that apply)

- A. It uses the Transformer decoder framework
- B. It retains masked multi-attention and feed forward layers
- C. It expanded to 12 layers with 117 million parameters
- D. It uses random initialization for positional encoding, updated during training
- E. It uses LSTM instead of Transformer

Q13 Which statements about BERT are CORRECT? (Select all that apply)

- A. Released in 2018
- B. It is a bidirectional model
- C. Trained on 3.3 billion words and 340 million parameters
- D. Can answer questions, predict statements, and translate text
- E. Uses the Transformer encoder architecture

Q14 Which statements about multi-task processing are CORRECT? (Select all that apply)

- A. Conventional models were used for single or few tasks
- B. A single model for multiple tasks can suffer from interference between different subsets
- C. Interference effect slows training and reduces generalization
- D. Foundation models aim to handle multiple domains, topics, languages, and modalities
- E. Multi-task processing is impossible with neural networks

Q15 Which statements about Mixture-of-Experts (MoE) are CORRECT? (Select all that apply)

- A. Combines multiple "expert" models
- B. Uses a gating mechanism for expert selection
- C. Improves capacity and efficiency of foundation models
- D. Can achieve same quality as dense models with less compute in pre-training
- E. All experts are always active for every input

Q16 Which models/algorithms were enabled by Transformer architecture? (Select all that apply)

- A. BERT
- B. GPT series
- C. ViT (Vision Transformer)
- D. LSTM-based models
- E. Traditional CNN-only models

True or False Questions (17-20)

Q17 Foundation models like GPT-4, GLM, and Stable Diffusion have billions of parameters, enabled by Transformer scalability and increased computing power.

- A. True
- B. False

Q18 GPT-1 uses the Transformer decoder framework, expanding the network scale to 12 layers, 768 attention dimensions, 12 attention heads, and 3072 feed forward hidden layers, totaling 117 million parameters.

- A. True
- B. False

Q19 BERT is a unidirectional model that analyzes text only from left to right, using the Transformer decoder architecture.

- A. True
- B. False

Q20 MoE (Mixture-of-Experts) uses a gating mechanism to select specific experts for each input, and can achieve effective pre-training with less computing resources than dense models require.

- A. True
- B. False

Fill in the Blank Questions (21-24)

Q21 Foundation models are trained with powerful _____ and _____ capabilities based on _____ amounts of data and _____ resources. They can quickly _____ to new tasks through _____.

Q22 GPT-1 uses the _____ of Transformer, retaining the _____ - layer and _____ layer. It expanded to _____ layers, _____ attention dimensions, _____ attention heads, and _____ feed forward hidden layers. The total parameters reached _____ million. For positional encoding, GPT-1 uses _____ similar to word vectors, updated during _____.

Q23 BERT is a _____ model released in _____, trained on _____ billion words with _____ million parameters. It uses the Transformer _____ architecture. BERT can _____ questions, _____ statements, and _____ text. The bidirectional nature means it analyzes _____ from both _____ before predictions.

Q24 MoE (-of-) combines multiple _____ with a _____ mechanism for _____ selection. Its advantages include improved _____ and _____, effective _____ with less _____, and ability to _____ up model or dataset under same _____ budget. The _____ effect in conventional models occurs when updating weights for one _____ affects another, slowing _____ and reducing _____.

Drag and Drop / Matching Questions (25-30)

Q25 Match each model to its Transformer component:

Model Component

- | | |
|-------------------------|-----------------------------|
| A. GPT-1 | 1. Transformer Encoder |
| B. BERT | 2. Transformer Decoder |
| C. Original Transformer | 3. Both Encoder and Decoder |

Write your answer as: A→?, B→?, C→?

Q26 Match each model to its key characteristic:

Model Characteristic

- | | |
|----------------------|---|
| A. GPT-1 | 1. Bidirectional, trained on 3.3B words |
| B. BERT | 2. Uses decoder, 117M parameters |
| C. Foundation models | 3. Massive scale, generalization, fine-tuning |

Write your answer as: A→?, B→?, C→?

Q27 Match each concept to the problem it addresses:

Concept Problem

- | | |
|------------------------|---|
| A. Fine-tuning | 1. Single model limited to one task |
| B. MoE | 2. Model updates for one scenario affect others |
| C. Interference effect | 3. Adapting foundation model to new domain |
| D. Multi-task model | 4. Scaling model capacity under compute budget |

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each scaling parameter to its GPT-1 value:

Parameter Value

- | | |
|-------------------------------|----------------|
| A. Layers | 1. 768 |
| B. Attention dimensions | 2. 12 |
| C. Attention heads | 3. 3072 |
| D. Feed forward hidden layers | 4. 117 million |
| E. Total parameters | 5. 12 |

Write your answer as: A→?, B→?, C→?, D→?, E→?

Q29 Match each era to its description:

Era Description

- | | |
|-------------------------|---|
| A. Pre-Transformer | 1. RNN and CNN era |
| B. Transformer era | 2. BERT, GPT, ViT with excellent performance |
| C. Foundation model era | 3. Billion-parameter models like GPT-4, LLaMA |

Write your answer as: A→?, B→?, C→?

Q30 Match each MoE component to its function:

Component Function

- | | |
|---------------------|--|
| A. Experts | 1. Decide which experts to use for each input |
| B. Gating mechanism | 2. Combine expert outputs for final prediction |
| C. Output mechanism | 3. Specialized sub-models for different inputs |

Write your answer as: A→?, B→?, C→?

Scenario-Based Questions (31-35)

Q31 A company needs to build a system that:

Generates coherent text (like GPT)

Understands context from both directions for question answering (like BERT)

Which architecture choices are MOST appropriate?

- A. Use GPT decoder for generation, BERT encoder for understanding
- B. Use only GPT for both tasks
- C. Use only BERT for both tasks
- D. Use traditional RNN for both tasks

Q32 A research lab has limited GPU resources but wants to train a large language model. They consider using MoE instead of a dense model.

Why might MoE be advantageous?

- A. MoE requires more compute than dense models
- B. MoE can achieve similar quality with less compute during pre-training
- C. MoE only works for small models
- D. MoE eliminates the need for any training data

Q33 A model is trained jointly on medical texts, legal documents, and news articles. The model performs poorly on all three compared to separate specialist models.

What is the likely cause and solution?

- A. The model is too small; solution: use MoE to increase capacity and reduce interference
- B. The data is too similar; solution: don't mix domains
- C. The learning rate is too high; solution: decrease learning rate
- D. The model is overfitting; solution: add more regularization

Q34 A researcher implements GPT-1 and notices that the model learns positional information differently than the original Transformer.

What is the difference and why?

- A. GPT-1 uses sine/cosine; original Transformer uses learned embeddings
- B. GPT-1 uses random initialization updated during training; original uses fixed sine/cosine
- C. Both use identical positional encoding
- D. GPT-1 doesn't use positional encoding

Q35 A foundation model trained on general internet text is fine-tuned on Huawei technical documentation to answer support questions.

What does this demonstrate about foundation models?

- A. Foundation models cannot adapt to new domains
- B. Foundation models can quickly adapt to new tasks and data through fine-tuning
- C. Fine-tuning always destroys previous knowledge
- D. The model must be retrained from scratch

Advanced Concept Questions (36-40)

Q36 GPT-1 modified the original Transformer decoder. Which change was NOT made?

- A. Increased layers from original to 12
- B. Changed positional encoding from sine/cosine to learned embeddings
- C. Added a new convolutional layer between attention and feed forward
- D. Increased attention dimensions to 768

Q37 BERT uses the Transformer encoder with bidirectional attention. What does bidirectional mean in practice?

- A. The model processes text twice: once left-to-right, once right-to-left
- B. Each token can attend to all other tokens in the sequence simultaneously
- C. Two separate models run in opposite directions
- D. The model only looks at previous tokens

Q38 In MoE, the gating mechanism outputs weights for each expert. If the gating gives 90% weight to Expert 3 and 10% to Expert 7 for a specific input:

- A. Only Expert 3 is used; Expert 7 is ignored
- B. Both experts contribute, but Expert 3 dominates the final prediction
- C. The model is broken
- D. All experts are used equally

Q39 Why do foundation models like GPT-4 have hundreds of billions of parameters while early models had millions?

- A. Because more parameters always guarantee better performance regardless of data
- B. Because Transformer scalability and increased compute enable larger models with better generalization
- C. Because smaller models are always worse
- D. Because parameters are free and unlimited

Q40 Which statement best describes the trajectory from Transformer to foundation models?

- A. Transformer enabled scalable architectures → larger models on more data → foundation models with strong generalization and adaptation
- B. Foundation models made Transformer obsolete
- C. RNNs are making a comeback
- D. CNNs will replace Transformers for all tasks

 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	B	Foundation models: massive data/compute, generation/generalization, fine-tuning
2	B	Transformer scalability enabled billion-parameter models
3	B	GPT-1 uses random init positional encoding, updated during training
4	A	GPT-1: 768 dims, 12 heads, 3072 FF, 117M params
5	B	Bidirectional = analyzes complete sequence from both directions
6	A	BERT = encoder (bidirectional), GPT = decoder (unidirectional)
7	B	Interference effect: updating one scenario affects others
8	B	MoE improves capacity/efficiency, less compute than dense
9	B	Model size key for performance; MoE scales under budget
10	B	Fine-tuning enables quick adaptation to new tasks
11	A, B, C, D	E is false (foundation models extend beyond NLP)
12	A, B, C, D	E is false (GPT-1 uses Transformer, not LSTM)
13	A, B, C, D, E	All statements about BERT are correct
14	A, B, C, D	E is false (multi-task processing is possible)
15	A, B, C, D	E is false (only selected experts are active)
16	A, B, C	D and E are false (LSTM and CNN-only predate Transformer)
17	A - True	Direct description from foundation models section
18	A - True	Direct description of GPT-1 architecture
19	B - False	BERT is bidirectional and uses encoder, not decoder
20	A - True	Direct description of MoE mechanism
21	generation, generalization, massive, computing, adapt, fine-tuning	Foundation model characteristics
22	decoder, framework, masked, multi, attention, feed, forward, 12, 768, 12, 3072, 117, random, initialization, training	GPT-1 architecture details
23	bidirectional, 2018, 3.3, 340, encoder, answer, predict, translate, context, directions	BERT characteristics
24	Mixture, Experts, experts, gating, expert, capacity, efficiency, pre-training, compute, scale, computational, interference, scenario, learning, generalization	MoE and multi-task concepts
25	A→2, B→1, C→3	GPT-1=decoder, BERT=encoder, Original=both
26	A→2, B→1, C→3	GPT-1=decoder 117M, BERT=bidirectional 3.3B words, Foundation=massive scale
27	A→3, B→4, C→2, D→1	Fine-tuning=adapt to domain, MoE=scale under budget, Interference=updates affect others, Multi-task=one model many tasks
28	A→5, B→1, C→2, D→3, E→4	Layers=12, Dims=768, Heads=12, FF=3072, Params=117M
29	A→1, B→2, C→3	Pre-Transformer=RNN/CNN, Transformer=BERT/GPT/ViT, Foundation=GPT-4/LLaMA
30	A→3, B→1, C→2	Experts=sub-models, Gating=select experts, Output=combine predictions
31	A	GPT decoder for generation, BERT encoder for bidirectional understanding
32	B	MoE achieves similar quality with less compute

33	A	Interference from mixing domains; MoE increases capacity
34	B	GPT-1: random init + learned; Original: fixed sine/cosine
35	B	Fine-tuning demonstrates quick adaptation capability
36	C	GPT-1 did NOT add convolutional layers
37	B	Bidirectional = each token attends to all others simultaneously
38	B	Both contribute, but Expert 3 dominates (90% weight)
39	B	Transformer scalability + more compute → larger models
40	A	Transformer → scalable → larger → foundation models